Digital Tax Administration and the Value of Frontline Discretion

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Abstract

Governments increasingly turn to technology to collect taxes more fairly and effectively. Yet replacing frontline officers also changes how citizens obtain assistance, contest demands and negotiate what they pay. Personal interaction can provide protection from administrative mistakes while creating opportunities for private advantage. What do taxpayers value in these relationships, and what would they prefer to replace with technology? I answer these questions through a conjoint experiment with 2,100 property owners in Lahore, Pakistan. When both inspector offers involve concessions, taxpayers are willing to pay approximately Rs 1,900 more to receive a correct bill from the outset rather than face an excessive initial demand followed by a negotiated reduction. Reaching the same final payment therefore does not eliminate their preference for being assessed correctly. However, the savings available through inspector concessions can reduce their willingness to choose accurate satellite assessment. When technology and a human inspector assess liability correctly at the same cost, with no informal payment under either arrangement, I find no statistically significant average preference for technology-based assessment. Perceived satellite fairness and confidence that errors in computer-generated bills can be corrected are associated with greater willingness to pay for satellite assessment. The fiscal implications depend on what personal intervention delivers: removing an excessive claim restores the intended liability, whereas concessions below that liability finance both taxpayer savings and informal payments. The findings suggest that accurate assessment as promised by digitalization alone may be insufficient to secure taxpayer support for replacing discretionary arrangements. For revenue authorities, the challenge is to make accurate decisions and legitimate correction credible while recognising that restricting private concessions also withdraws financial benefits from taxpayers.

JEL: H26, H71, D73, O33

1 Introduction

A central problem in public finance is how governments can raise funds for public services and redistribute resources while limiting their economic and administrative costs. The capacity of a state to raise revenue depends on the institutions that translate statutory rules into effective taxation, as well as on economic development and political choices (Besley and Persson, 2014). Research establishes the role of deterrence in tax compliance (Slemrod, 2019). Taxpayers' perceptions of the authority also influence compliance (Luttmer and Singhal, 2014). Murphy (2004) explores how trust plays an important role in the relationship between the tax authorities and taxpayers accused of tax avoidance, while van Dijke and Verboon (2010) find that procedural fairness is particularly consequential for voluntary compliance among citizens with low trust in authorities. Similarly, Kirchler et al. (2008) provide a formal framework which distinguishes between trust-based cooperation and enforced compliance. These contributions establish the relevance of the taxpayer–authority relationship to tax collection.

In many low-middle-income countries, frontline-led tax administration bundles functions that taxpayers may evaluate and trust differently. An inspector can establish taxable facts, explain a liability, and help correct a record. The same discretion can also enable preferential treatment in exchange for an informal payment. A taxpayer who distrusts an inspector to assess a property fairly may nevertheless value access to that inspector when a bill is disputed. The distinction has direct fiscal significance. Correcting an excessive assessment improves the application of the intended tax rules, whereas negotiating a payment below the correct liability benefits the taxpayer at the expense of public revenue. Both can make personal interaction with the officer attractive to the taxpayer, but administratively, they imply different reasons for preserving or restricting official discretion.

These distinctions become particularly consequential as revenue authorities are increasingly introducing technologies that change the role of frontline officials. Technology shifts decisions from "street-level" towards "system-level" bureaucracy, where algorithms apply rules to recorded information (Bovens and Zouridis, 2002). This reallocation can substitute for or complement frontline officers' work (Autor et al., 2003; Bullock, 2019). Routine tasks governed by explicit rules, such as applying a statutory calculation to a recorded tax base, are more amenable to automation (Bullock, 2019). However, complex tasks involving communication and problem-solving may still need officers' judgement and local knowledge (Gräfe et al., 2024). In taxation, this may mean that reducing personal contact through technology can limit officer rents, but it can also change access to more complex officer tasks that citizens previously relied on to navigate the system. Understanding demand for digitalisation of tax administration therefore requires distinguishing taxpayer preferences over the assessor of their tax liability from preferences over assessment outcomes, financial transactions, and access to recourse.

Understanding these taxpayer preferences is crucial because of the scale of investment in digitalization of tax administration. Among 116 countries with World Bank taxation-related projects approved between 2010 and 2022, 91 had a project containing a tax-modernization or

information-technology component (Okunogbe and Tourek, 2024). Evidence on the effectiveness of digitizing tax administration in low- and middle-income countries shows mixed results, and the returns from digital reforms depend on the surrounding administrative arrangements. In Tajikistan, electronic filing reduces compliance time, raises payments among firms previously more likely to evade, and reduces opportunities for extortion (Okunogbe and Pouliquen, 2022). However, in Ecuador, firms offset third-party revenue corrections through higher reported costs, limiting the increase in collection (Carrillo et al., 2017). In Ghana, technology complements collectors' work by changing how they allocate time and learn about taxpayers (Dzansi et al., 2022). Similarly, the distinction between administrative improvement and collection is important. Electronic filing and payment in Senegal improve the completeness of administrative information while producing limited measured compliance gains; digital payments in Eswatini improve payment outcomes, with effects that differ across taxpayers and adoption routes (Czajka et al., 2026; Santoro et al., 2026). In property taxation, digital records and valuation systems operate within a chain of billing, collection and enforcement (Prichard and Fish, 2017). This paper examines the taxpayer interests within that chain: the value of accurate administration and the private benefits that can accompany discretionary settlement.

I study these preferences for urban property-tax administration in Lahore, the capital of Punjab, Pakistan. Inspectors play a central role in establishing the property characteristics used to calculate liabilities, communicating with taxpayers, and initiating amendments to the administrative record. Their recommendations require supervisory approval, but their responsibility for producing property-specific information makes them central to taxpayers' encounters with the authority. Lahore also provides a relevant reform environment: Punjab has introduced GIS-based administrative changes and shifted property valuation from annual rental value towards capital value based on administrative rates. Among taxpayers in my survey, approximately half report little or no trust in officers to value properties fairly. This dissatisfaction makes alternatives to inspector assessment salient. However, it remains unclear whether taxpayers favour replacing tax officers entirely with technology-based administration and whether they expect errors under such systems to be readily corrected.

How do taxpayers value digital assessment relative to the administrative services and private financial benefits available through frontline discretion? I investigate this question through three related comparisons. First, do taxpayers prefer an inspector or a satellite-based system when both produce the correct bill at the same cost? Second, how do initial tax-bill errors and the terms of concessions in exchange for informal payments shape choices over assessment arrangements? Third, how do these choices relate to trust in officers' valuation fairness and expectations about correcting computer-generated errors? These comparisons connect the design of tax administration to the particular outcomes and institutional expectations underlying taxpayers' choices.

I study responses of 2,100 property taxpayers recruited from an administrative records-based sampling frame in Lahore, combining survey measures with a choice experiment. In the experiment, taxpayers repeatedly choose between assessment receipts displaying the assessor, a

correct reference liability, the initial bill, any informal payment, the final official tax payment, and total expenditure. Inspector offers can include a concession in exchange for an informal payment; satellite offers do not. Each of the 2,100 respondents completes four fixed tasks followed by six Bayesian adaptive tasks, giving 21,000 paired choices. The adaptive tasks use earlier responses to select subsequent comparisons. I estimate preferences conditional on the sequential design. Survey measures capture prior departmental contact, officer trust, expected assessment accuracy and impartiality, expected correctability, and ownership and documentation. A separate offer of computer-estimated information about respondents' own properties provides evidence on demand for independent property information.

Three sets of findings emerge. First, the full choice experiment provides no statistically significant evidence of an average premium for satellite assessment over an inspector, despite the 59.8% satellite share in the opening fixed-order comparison. Taxpayers nevertheless value correct initial bills even when final expenditure is unchanged, and their valuation of inspector concessions varies with the initial assessment. Second, accurate assessment competes with the private savings available through inspectors. In model-implied comparisons with a 10% net inspector saving, providing a correct satellite bill rather than an excessive one raises satellite selection by 14.5 percentage points. The associated fiscal accounting distinguishes the removal of excessive claims from concessions below correct liability, which finance both taxpayer savings and informal payments. Third, institutional expectations distinguish preferences over these arrangements: perceived satellite fairness is positively associated with satellite valuation, while expected access to correction is associated with higher valuations of both satellite assessment and inspector concessions. Together, the findings show how demand for accurate administration can coexist with a financial interest in discretionary settlement. The challenge for revenue authorities is to make correct treatment credible and accessible while limiting privately negotiated departures from tax rules.

The paper contributes to research on taxpayer cooperation by connecting specific institutional beliefs to choices over tax administration. The literature on tax morale and trust establishes the importance of taxpayers' relationship with the state (Luttmer and Singhal, 2014; Kirchler et al., 2008), while subsequent work distinguishes forms of trust and examines how interactions with officials shape taxpayer perceptions (Gangl et al., 2015; Mascagni et al., 2025). I examine a related but distinct question: how taxpayers choose between assessment arrangements when the assessor, initial assessment, and financial terms are made explicit. This approach distinguishes confidence in officers' fairness from beliefs about correcting computer errors, and relates both to assessor choice. The explicit attributes show how dissatisfaction with an authority can coexist with demand for some of its services or settlements.

The paper also contributes to public-finance research on discretion, collector incentives, and digital tax administration. Khan et al. (2016a) examine how incentives facing collectors change revenue collection and informal payments. Okunogbe and Pouliquen (2022) study how replacing in-person filing changes compliance costs and tax payments, while Knebelmann et al. (2026) examine the accuracy, equity, and acceptability of algorithmic property valuation. I comple-

ment this work by separating taxpayer responses to assessor identity, initial tax-bill errors, and concessions in exchange for informal payments within a common choice framework, and by examining their relationship with expected recourse. The distinction between private attractiveness and fiscal desirability is central: an arrangement can appeal to a taxpayer because it supports accurate taxation or because it permits a favourable departure from the rules.

Political-economy evidence from Bangladesh describes businesses supporting administrative simplification while defending informal financial advantages (Hassan and Prichard, 2013). The contribution here is to quantify the trade-offs within taxpayers using an explicit correct-liability benchmark. That benchmark distinguishes removal of an excessive demand from a concession below the stated liability, while the full sequence varies assessor identity, initial accuracy and settlement terms. The resulting valuations give the taxpayer side of administrative reform a measurable private and fiscal content.

For revenue authorities, the findings make the arrangements surrounding assessment a salient part of technological reform. Taxpayers may evaluate a digital system partly through their expectations about what happens when its decisions are wrong. At the same time, acceptance of a private concession cannot be treated as evidence that discretionary bargaining should be preserved. The relevant design question is how to provide dependable assessment and accessible correction while limiting preferential treatment. This paper supplies evidence on taxpayers' choices and the beliefs associated with them; establishing how alternative arrangements affect compliance, revenue, and welfare requires evidence from their implementation.

The paper proceeds from the institutional setting in Section 2 to the taxpayer's decision environment in Section 3. Section 4 describes the sample and experiment. Section 5 defines the estimands and estimation methods. Section 6 develops the evidence from common comparisons to full-sequence trade-offs and their institutional and fiscal significance. Section 7 discusses the implications for digital tax administration.

2 Tax Administration and Frontline Discretion

2.1 Tax-Liability Determination and Frontline Officers

Punjab's Urban Immovable Property Tax (UIPT) is administered by the provincial Excise and Taxation Department. The department divides the urban jurisdictions subject to UIPT into a four-level geographical hierarchy, illustrated in Figure B.1. The primary unit for property inspection, assessment, collection, and enforcement is called the tax "circle". Each circle is assigned an inspector, supported by a clerk and a constable. The inspector leads property visits, determines the relevant property characteristics, and takes primary responsibility for the recommended assessment. The clerk and constable assist with measurement, documentation, record preparation, and the delivery of notices.

The circle team cannot enter its recommendation directly into the official tax record. It must submit the assessment to the Excise and Taxation Officer (ETO), whose approval is mandatory. The ETO reviews the measurements, classifications, and supporting record prepared by the inspector-led team. The process therefore separates informational control from formal authority: the inspector controls the production of property-specific information, while the ETO controls whether the recommendation receives formal approval. ETO is not a party to any informal bargain between the inspector and the taxpayer. ETO approval nevertheless matters because the inspector must produce a record that is sufficiently complete and internally consistent to obtain formal approval.

Survey evidence illustrates the range of interactions between taxpayers and the department. In a frontline tax officer survey, inspectors and constables report an average of **[MEAN]** citizen contacts per week. In a separate taxpayer survey conducted in March–April 2026, 39.3% reported that they or another household member had contacted E&T regarding their property during the preceding year. Among the respondents reporting contact, 45.9% mentioned paying a tax bill, 45.9% seeking general information, and 10.2% correcting property records. These interactions therefore extend beyond payment to obtaining information and seeking amendments to the records underlying tax liability.

2.2 The Taxpayer–Administration Relationship

A taxpayer survey conducted in the same study site describes the taxpayer–administration relationship in Lahore.[1] It distinguishes three related but conceptually separate dimensions of the fiscal relationship: perceived diversion during collection, trust in the institutions administering the tax and perceptions of the return from taxation. These distinctions matter because dissatisfaction with one dimension does not reveal which features of frontline-led tax administration taxpayers would prefer to retain or remove.

Table 1 reports these measures. The attribution question divides the perceived collection gap between informal payments and collection difficulties. On average, citizens attribute 62.2% of that gap to informal payments, and 71.0% assign more than half of the gap to this explanation. These responses indicate a widespread perception that private transactions divert revenue from the formal collection process.

[1]The survey was conducted as a joint work for the paper, "Abbas et. al., 2026. The Political Economy of Progressive Tax Reform: Experimental Evidence from Pakistan". See AEA registry here

Table 1: Baseline surveys: perceptions of collection, trust and public services

Measure	Count / valid responses	Percent
Panel A. Perceived diversion from collection		
Mean collection gap attributed to informal payments	4,897 responses	62.2
Attribute more than half the gap to informal payments	3,476 / 4,897	71.0
Panel B. Trust in officials, government and other citizens		
No trust at all in property-tax staff	1,831 / 4,897	37.4
No trust at all in government	1,699 / 4,897	34.7
No trust at all in other Lahore citizens	738 / 4,897	15.1
Panel C. Perceived return from taxation		
Disagree that Punjab uses tax revenue for people like them	3,147 / 4,897	64.3

Notes: Citizen baseline survey from the same study site, Phase 1. The attribution question allocates the collection gap between informal payments and collection difficulties; its mean is a perceived share. Trust uses only the unambiguous no-trust endpoint. The service item combines strongly disagree and disagree to some extent. These are separate measures; no index is constructed.

Distrust extends beyond the collection process itself. No trust at all in property-tax staff is reported by 1,831 respondents, or 37.4%, while 1,699 respondents, or 34.7%, report no trust at all in government. By comparison, 15.1% report no trust at all in other Lahore citizens. The pattern therefore points to a particular concern with the fiscal relationship rather than a general unwillingness to trust others. A separate question captures perceptions of the fiscal return: 3,147 respondents, or 64.3%, disagree that the Government of Punjab uses tax revenue to provide goods and services for people like them.

Similarly, a survey of frontline officers provided a complementary view of the same collection environment. Among 292 officials ranking five explanations for incomplete collection, weak penalties and poor public services receive the most first-place rankings. Taxpayers offering informal payments are ranked first by 9.6% of officials for low-value property taxpayers and 8.2% when asked about the medium- and high-value property taxpayers.

2.3 Frontline-led Tax Administration and Digitalisation

A taxpayer who distrusts the administration may nevertheless value access to an inspector when an assessment is wrong. Property-tax liability depends on recorded property characteristics as well as the valuation rule, so a correctly applied formula can still produce an incorrect bill. Frontline officials can inspect the property, examine supporting documents and initiate authorised corrections.

Such assistance is distinct from the financial advantages available through informal accommodation. Piracha and Moore (2015, 2016) describe how inspectors' local knowledge, continuing relationships with property taxpayers and discretion over assessment and enforcement sustain arrangements that can benefit both parties while limiting public revenue. Frontline-led tax administration can therefore provide access to lawful correction as well as concessions in exchange for informal payments. I distinguish *non-appeal administrative correction*, an autho-

rised amendment within the assessment chain, from *formal appeal*, which follows the statutory appeal route. An *inspector concession in exchange for an informal payment*, by contrast, reduces the official demand in return for a private payment to the inspector.

Taxpayers' digital comfort and access to documents are relevant to how they navigate these arrangements. In the survey of 2,100 property owners, 36.2% of the 2,081 respondents providing a substantive answer report little or no comfort completing an online form. Separately, 64.9% of 2,070 substantive respondents report at least some difficulty locating documents needed for property-related or other official work. These reported constraints suggest that assistance may remain valuable even when property measurement or liability calculation is automated.

This combination of assistance, discretion and taxpayer constraints is central to understanding how digitalisation may change the role of frontline officials. Punjab's earlier GIS reforms sought to reduce inspectors' discretion, while the 2024 reform shifted valuation from annual rental value to capital value based on administrative rates (World Bank, 2020; Government of Punjab, 2024).

My interviews with senior bureaucrats motivated the experimental comparison between inspector and satellite-based assessment. The experiment examines taxpayers' stated choices between hypothetical assessment arrangements when both the correctness of the initial bill and the financial terms of personal interaction are explicit; it does not evaluate the implemented valuation reform. Section 3 formalises these trade-offs in a model that distinguishes preferences over the assessor from willingness to accept an inspector concession in exchange for an informal payment under correct, excessive or understated initial bills and different payment terms.

3 Conceptual Framework

Property taxation brings an owner into a relationship with the administration that extends from establishing a liability to settling the bill. An inspector can establish taxable facts, explain the demand and initiate corrections. Discretion can also allow a lower official payment in exchange for an informal payment. Research on tax administration shows how dealings between taxpayers and officials can shape both official receipts and private payments (Khan et al., 2016b; Okunogbe and Pouliquen, 2022). The framework examines how taxpayers value the initial demand, the assessor and the settlement alongside their final expenditure.

3.1 The Taxpayer's Assessment and Payment Experience

Consider a property owner i who receives an initial bill I_{ij} under administrative arrangement j. The amount demanded need not equal the liability implied by the tax rules. In practice, the owner may be uncertain about that liability and may rely on the administration to explain or verify the demand.

Reference liability. The experiment states the correct liability Q_i on both receipts. The framework assumes that taxpayers understand and use this amount when comparing offers.

The initial bill may remain unchanged, or it may be reduced through two distinct channels. First, an authorized correction after a taxpayer's appeal can remove an excessive demand. Second, an inspector may also offer a lower official payment in exchange for an informal payment. The informal payment offer can remove the initial excess, reduce the payment below the correct liability, or do both.

Let T_{ij} denote the final official payment, $B_{ij} \geq 0$ the informal payment, and $A_{ij} \geq 0$ the cost incurred when appealing, set to zero when there is no appeal. If appeal and an informal deal are alternative routes, total taxpayer expenditure is

$$p_{ij} = \begin{cases} T_{ij} & \text{without an appeal or informal payment,} \\ T_{ij} + A_{ij} & \text{with an appeal,} \\ T_{ij} + B_{ij} & \text{with an informal payment.} \end{cases} \tag{1}$$

Equation 1 emphasizes that a taxpayer can only reduce his total tax expenditure by appealing or paying an informal payment if the reduction in final official payment is greater than the appeal cost or the informal payment made to obtain it.

3.2 What Matters Beyond the Amount Paid?

The owner may judge the experience by more than final expenditure. Consider two inspector concessions with initial bills of Rs 20,000 and Rs 22,000. Both end with an official payment of Rs 16,000 and an informal payment of Rs 3,000. Total expenditure is Rs 19,000 under either offer, but only the first initial bill is correct. Allowing this distinction follows research on the valuation of procedures as well as outcomes (Frey et al., 2004; Bolton et al., 2005).

Whether the initial bill is correct. Compare the initial bill, I_{ij}, with the stated correct bill, Q_i:

$$e_{ij} = \begin{cases} 0 & \text{if } I_{ij} = Q_i \quad \text{(correct bill),} \\ O & \text{if } I_{ij} > Q_i \quad \text{(bill too high),} \\ U & \text{if } I_{ij} < Q_i \quad \text{(bill too low).} \end{cases}$$

Even when total expenditure is the same, the taxpayer may care whether the initial bill was correct. Let $h_i(e)$ represent this valuation, taking a correct bill as the reference:

$$h_i(e) = \begin{cases} 0 & \text{for a correct initial bill,} \\ h_i(O) & \text{for an initial bill that is too high,} \\ h_i(U) & \text{for an initial bill that is too low.} \end{cases}$$

These values represent a bill assessed by an inspector, with no informal payment.

Whether an informal payment is involved. Let $D_{ij} = 1$ when an informal payment is made and $D_{ij} = 0$ otherwise. The term $m_i(e)$ measures the taxpayer's reluctance to accept an inspector's informal deal, compared with an inspector arrangement without an informal payment, at the same initial bill and total expenditure.

A positive $m_i(e)$ means the taxpayer dislikes the deal; a negative value means the taxpayer prefers it. This valuation may depend on whether the initial demand was correct, too high, or too low. Allowing that dependence recognises that people can judge the same type of transaction differently across circumstances (Barr and Serra, 2009; Banerjee, 2016)

Assessor choice in the experiment. The experiment asks the taxpayer to compare bills assessed by an inspector with bills assessed through a satellite-based system. In these hypothetical scenarios, satellite assessment never involves an informal payment. Inspector assessment may involve one, depending on the offer presented. Write

$$S_{ij} = \begin{cases} 1 & \text{for satellite assessment,} \\ 0 & \text{for inspector assessment.} \end{cases}$$

Let $\beta_i(e)$ measure how much the taxpayer values satellite assessment relative to inspector assessment when neither involves an informal payment, the initial bill is the same, and total expenditure is equal. A positive value indicates a preference for satellite assessment; a negative value indicates a preference for the inspector.

With $S_{ij} = 1$ for satellite assessment and $D_{ij} = 1$ for a positive informal payment, the owner's evaluation is

$$V_{ij} = -p_{ij} + h_i(e_{ij}) + \beta_i(e_{ij})S_{ij} - m_i(e_{ij})D_{ij}. \tag{2}$$

Maintained assumption: Satellite offers have $D_{ij} = 0$.

Expenditure enters linearly, so the other terms are expressed in rupees. The informal payment's cash cost is already included in p_{ij}. For the same initial bill, the model treats two informal deals as equally attractive if the taxpayer pays the same total amount, even if one deal involves a larger informal payment and a smaller official tax payment. It allows taxpayers to care about making an informal payment, but does not separately capture how they feel about its size.

3.3 Choice and Experimental Comparisons

The owner chooses the available arrangement with the highest payoff. Let H_0 denote an inspector offer without an informal payment and H_1 an inspector offer with a concession for an informal payment. With the same initial-bill error e, define the expenditure saving as $s_i = p_{iH_0} - p_{iH_1}$.

The concession is weakly preferred when

$$s_i \geq m_i(e). \tag{3}$$

Against a satellite offer with the same initial bill, define $s_i^S = p_{iS} - p_{iH_1}$. The corresponding condition is

$$s_i^S \geq \beta_i(e) + m_i(e). \tag{4}$$

An owner can therefore prefer satellite assessment on equal terms yet accept an inspector concession when its saving is sufficiently attractive.

These comparisons organise the experiment. Equal correct bills and expenditure allow assessor preference to be examined. Comparisons of initially correct and incorrect bills examine whether the initial demand matters beyond eventual expenditure. Inspector offers with and without concessions examine whether savings alone explain acceptance and whether transaction valuation changes with the initial bill. The full sequence examines how these valuations combine within taxpayers.

Equal expenditure does not remove every other difference between offers. When two inspector concessions begin with different bills, their comparison combines the valuation of the initial demand with any change in transaction valuation. Separating these components requires the wider receipt variation and the maintained model. Appendix A derives these comparisons and states the empirical restrictions. The framework permits, but does not impose, greater attraction to inspectors or their concessions following overassessment.

The offered terms are taken as given. The framework does not model bargaining or separately identify the value of explanation, assistance or authorised correction. Beliefs about these services may contribute to assessor valuation; their association with choices can be examined using the survey measures.

3.4 Private Savings and Official Receipts

The owner's financial saving need not equal the reduction in official receipts. Define saving relative to paying the correct liability as $G_{ij} = Q_i - p_{ij}$. Then

$$Q_i - T_{ij} = B_{ij} + A_{ij} + G_{ij}. \tag{5}$$

The shortfall in official payment equals the informal payment, any appeal cost and the taxpayer's net saving. Experimental receipts implement the no-appeal and informal-payment branches of equation (1), with $A_{ij} = 0$ and $p_{ij} = T_{ij} + B_{ij}$. The saving is positive only when total expenditure falls below Q_i. Removing an initially excessive demand does not itself create a shortfall relative to the correct liability. This distinction connects the valuation of settlements to the fiscal accounting of selected offers.

4 Data and Experimental Design

4.1 Study Site and Survey Procedures

The study is based on a taxpayer survey of 2,100 completed interviews conducted between July and August 2026. The study site is Lahore, the capital of Punjab, Pakistan. With a population of 13 million people, Lahore is the world's 15th largest city. Permissions to conduct a survey were sought from district government and Punjab's Excise and Taxation (E&T) Department, a provincial government revenue authority that administers the collection and billing of the primary property tax in metropolitan cities. E&T administers the taxation of almost 1 million properties in Lahore.

The consent protocol explained voluntary participation, confidentiality and the hypothetical nature of the choices. Survey respondents were offered Rs 2,000 in mobile credit as time compensation, from which the optional information price could be deducted. Ethical approval was obtained from the University of Sussex Social Sciences and Arts Cross-School Research Ethics Committee (C-REC; reference ER/AF619/2).

4.2 Sample and recruitment

Table 2 summarizes recruitment from a frame of 10,564 properties. The survey used stratified random sampling, with strata defined by locality, land area, and number of storeys. Interview targets were allocated across localities in proportion to aggregate recorded covered area and across land-area-by-storey strata within each locality in proportion to eligible property counts. Properties were randomly selected without replacement within strata, with reserve samples drawn for replacement recruitment (Appendix B). Of the 6,088 properties approached, 2,100 yielded completed owner interviews, giving a completion rate of 34.5%. All approached properties link to the administrative panel of 23,233 properties, allowing comparisons of observable characteristics between respondents and respondents who refused the survey.

Table 2: Survey recruitment and response rate

Recruitment stage	Count or rate
Remote-sensing sampling frame	10,564
Properties approached	6,088
Completed interviews	2,100
Without a completed interview	3,988
Properties not approached	4,476
Completion rate among approached properties	34.5%

Notes: Counts refer to unique properties, with one completed owner interview per interviewed property. The completion rate is the number of completed interviews divided by the number of properties approached. All 6,088 approached properties link to the administrative panel, which contains 23,233 usable property identifiers.

Table 3 describs 69.1% occupy plots of ten marla or less, while 13.6% exceed twenty marla. Properties average 2.82 storeys and approximately 6,250 square feet of covered area. Enumerator ratings place perceived wealth and education above the midpoint of their five-point scales, averaging 3.76 and 3.51, respectively, while perceived influence averages 2.99. These observations complement the observed asset profile: air conditioners were present at 81.4% of properties, air coolers at 35.7%, cars or trucks at 64.3%, and motorcycles or scooters at 71.1%. Relative to Lahore district in 2024–25, motorcycle prevalence is similar (73.1%), but cars or trucks are considerably more common than the broader category of cars, trucks, buses, or vans in the district benchmark (17.3%) (Bureau of Statistics, Government of the Punjab and UNICEF Punjab, 2025). Air-conditioning prevalence also exceeds the older Lahore estimate of 20.8% in 2011 (Bureau of Statistics Punjab, 2024).

Prior engagement with the tax administration is common: 57.9% report household contact during the preceding year. Nevertheless, the stated likelihood of seeking a record check or correction averages only 1.93 out of ten, indicating limited intended use of this procedure among those providing a substantive answer. The survey also includes social-desirability items to account for respondents' tendency to present themselves favourably. While 62.2% report always being courteous, 48.8% report always admitting mistakes, and 53.4% acknowledge resentment when they do not get their way.

Table 3: Respondent and property characteristics

Characteristic	Valid responses	Mean or %	Standard deviation
A. Enumerator assessments			
Perceived wealth (1–5)	2,100	3.76	0.79
Perceived influence (1–5)	2,100	2.99	1.05
Perceived education (1–5)	2,100	3.51	0.96
B. Assets observed at the property			
Air cooler observed (%)	2,100	35.7	
Air conditioner observed (%)	2,100	81.4	
Motorcycle or scooter observed (%)	2,100	71.1	
Car or truck observed (%)	2,100	64.3	
C. Property characteristics			
Land: $\leq$ 5 marla (%)	2100	35.5	
Land: > 5 to 10 marla (%)	2100	33.6	
Land: > 10 to 20 marla (%)	2100	17.3	
Land: > 20 marla (%)	2100	13.6	
Storeys	2,100	2.82	0.47
Covered area (sq.ft.)	2,100	6249.74	10581.10
D. Social-desirability items			
Always courteous, even to disagreeable people (%)	2,098	62.2	
Always willing to admit a mistake (%)	2,083	48.8	
Sometimes resentful when not getting one's way (%)	2,092	53.4	
E. Tax administration: contact and intentions			
Household contact with tax department, past year (%)	2,080	57.9	
Likelihood of seeking record checking/correction (0–10)	1,385	1.93	2.46

Notes: The sample comprises 2,100 completed interviews. Statistics are unweighted. Valid responses are counted separately for each row. Rows marked % report percentages; other rows report means and standard deviations.

4.3 The Tax-Assessment Choice Experiment

The paper's main experiment elicits taxpayers' preferences over tax-assessment and payment arrangements. Each of the 2,100 respondents completed ten choices between hypothetical tax receipts. Before the experiment, each respondent was thoroughly debriefed about every attribute. Each receipt described an assessment and its financial consequences. It displayed the assessor, the correct reference liability, the initial bill, any informal payment, the final official tax payment, and total expenditure. Respondents selected their preferred receipt from each pair. Appendix Figure C.1 shows an illustrative pair.

The intended sequence comprised four fixed comparisons followed by six Bayesian adaptive comparisons. The fixed comparisons establish how respondents choose under specific assessment and payment conditions. The adaptive comparisons introduce further variation in the terms offered, allowing the analysis to estimate how taxpayers trade administrative features against expenditure.

The experiment builds on stated-choice methods that elicit preferences by presenting alternatives with explicit monetary and nonmonetary attributes (Mas and Pallais, 2017a; Wiswall and Zafar, 2018; Maestas et al., 2023b). Related conjoint research uses comparisons between tax-policy packages to examine how their constituent features shape public support (Ballard-Rosa et al., 2017). Here, the object of choice is the taxpayer's own assessment and payment arrangement.

4.3.1 Experimental Attributes and Choice Task

The experiment varies three features of tax administration. The first is *who assesses the bill*: a tax inspector or a satellite-based system. Satellite offers never involve an informal payment; inspector offers may or may not involve one. Including inspector offers without informal payments makes it possible to compare the two assessors without simultaneously introducing a private transaction.

The second feature is *whether the initial bill is correct*. Both receipts state the same reference liability for the comparison, while an initial bill can equal, exceed, or fall below that amount. Stating the correct liability gives respondents a common benchmark for evaluating the assessment. It also distinguishes a reduction that removes an excessive demand from one that lowers the official payment below the amount owed. Each choice task specifies a reference liability: the amount legally due in the scenario presented to the respondent. For respondents exempted from property tax, actual legal liability is zero. These respondents were instead shown a hypothetical positive liability determined by their survey sampling cell, defined by land-area and storey bins. This amount serves as the correct tax liability within the experimental scenario, rather than an amount they actually owe.

The third feature is *the payment arrangement*. An inspector may offer a reduction in the official bill in exchange for an informal payment. The receipts show these amounts separately and report their sum as total expenditure. This presentation distinguishes the reduction in the government's claim from the taxpayer's financial saving: part of the reduction may instead be paid to the inspector. Variation in expenditure provides the monetary comparison used to value the administrative features of the offers.

Receipt amounts are scaled to the respondent's reference liability. They therefore represent common types of assessment and settlement rather than identical rupee amounts for every respondent. The attributes are subject to the accounting and institutional restrictions of the exercise: expenditure equals official plus informal payment, and informal payments occur only under inspector assessment. The choices are hypothetical and do not alter respondents' actual tax liabilities.

4.3.2 Fixed Comparisons: Assessor Choice, Informal Settlement, and Initial Accuracy

The four opening comparisons examine distinct aspects of the taxpayer's assessment experience.

Assessor choice at equal financial terms. Figure C.2 shows the first comparison, which presents a satellite assessment and an inspector assessment. Both initial bills are correct, total expenditure is equal, and neither receipt includes an informal payment. This comparison measures stated assessor choice when the displayed assessment and financial outcomes coincide. It establishes the starting point for asking whether a preference for satellite assessment coexists with acceptance of a financial concession from an inspector.

Acceptance of an informal-payment concession. Figure C.3 shows the second comparison between two inspector assessments, both beginning with the correct bill. One requires payment of the correct liability without an informal payment. The other offers a lower official payment in exchange for an informal payment. Holding the assessor fixed focuses the choice on the settlement and its financial consequences. Together, the first two responses show whether the same taxpayer chooses satellite assessment on equal terms yet accepts an inspector's concession when the available alternatives change. Acceptance alone does not establish a favourable view of informal payments, because the concession also changes expenditure.

Initial accuracy at comparable final expenditure. Figures C.4 and C.5 present third and fourth comparisons respectively, which show two inspector offers involving informal payments, one beginning with a correct bill and the other with an excessive bill. The third permits different informal-payment amounts while keeping the total payment almost identical. The fourth holds both total expenditure and the informal payment exactly equal. These comparisons examine whether taxpayers distinguish a correct initial demand from an excessive demand followed by a reduction, even when the eventual financial burden is similar or identical.

Because both alternatives in these latter comparisons involve informal settlement, their interpretation includes how the initial demand affects the evaluation of that settlement. Separating the valuation of initial accuracy from the valuation of an informal transaction requires the wider receipt variation and the model introduced in Section 5.

4.3.3 Adaptive Choice Design

The remaining six comparisons used earlier answers to select subsequent offers. The purpose was to learn how strongly taxpayers value the features of an assessment arrangement. Choosing satellite assessment when both receipts imply the same expenditure, for example, reveals a preference under those conditions. Comparing the assessors at different expenditure levels provided additional information about how much the taxpayer was willing to pay for that preference. The adaptive procedure followed the Bayesian approach developed by Drake et al. (2025a).

The adaptive-design specification represents taxpayer i's evaluation of receipt j as

$$u_{ij} = -c_{ij} + \beta_i S_{ij} - m_{0i} D_{ij} + m_{1i} D_{ij} o_{ij}. \tag{6}$$

Here, S_{ij} indicates satellite assessment and D_{ij} indicates a positive informal payment. Total expenditure is expressed as a percentage of the respondent's reference liability:

$$c_{ij} = 100\frac{T_{ij} + B_{ij}}{Q_i}. \tag{7}$$

The variable o_{ij} measures initial overassessment, with zero corresponding to a correct or understated bill and one corresponding to an initial bill 15 percent above the reference liability:

$$o_{ij} = \frac{\max\{100(I_{ij} - Q_i)/Q_i,\, 0\}}{15}. \tag{8}$$

The expenditure coefficient was normalised to -1, so β_i, m_{0i} and m_{1i} were measured in percentage points of the reference liability. The parameter β_i captures the value of satellite assessment relative to inspector assessment without an informal payment, holding expenditure constant. The parameter m_{0i} captures the cost of entering an informal-payment transaction when the initial bill is correct. The interaction allows that cost to change when the initial bill is excessive: at 15 percent overassessment, it equals $m_{0i} - m_{1i}$. A positive m_{1i} therefore makes the informal transaction more attractive following overassessment. The financial cost of the informal payment is already included in expenditure.

To allow for inconsistent choices, the design assigns the following probability to selecting receipt B rather than receipt A:

$$\Pr(\text{B chosen}) = \frac{\exp\{\mu_i(u_{iB} - u_{iA})\}}{1 + \exp\{\mu_i(u_{iB} - u_{iA})\}}, \tag{9}$$

where $\mu_i > 0$ governs how closely choices follow the utility difference. Higher values imply more consistent selection of the receipt with greater utility.

The procedure begins with a prior distribution over these parameters. After each answer, Bayes' rule increases the weight on parameter values that make the observed choice more likely. This updated distribution is then used to evaluate possible subsequent receipt pairs. The selection criterion favours the pair expected to provide the most additional information about the parameters. Thus, later comparisons depend on what earlier answers have already revealed. Appendix C reports the priors, updating equation, information criterion and available implementation evidence.

The adaptive stage included comparisons between two inspector offers, between two satellite offers, and across assessors. Inspector pairs vary assessment and settlement terms while holding assessor identity fixed. Cross-assessor pairs examine how institutional choice responds when initial bills and financial terms also differ. Satellite pairs involve no informal payments, although their initial errors and expenditure vary together.

The fixed comparisons provide choice shares under specified conditions; the full sequence supports estimation of preferences and their monetary equivalents. The design utility specifies how candidate comparisons are evaluated during adaptive selection. The empirical model in

Section 5 allows a richer valuation of initial assessment errors and is estimated from the receipts actually presented and the choices recorded.

4.4 Supplementary Survey Measures and Property-Information Offer

Alongside the choice experiment, the survey elicited respondents' experience with tax officials, expectations about satellite assessment, and ownership and documentation status. These measures are used to examine heterogeneity in experimental preferences. They were collected before the choice tasks; the separate property-information offer described below followed them. Appendix B.2 provides the full questionnaire and variable definitions.

Experience with and perceptions of tax officials. Respondents were asked whether they or another household member had contacted the Excise and Taxation Department about their property during the preceding year. Those reporting contact answered further questions about the interaction, including whether an informal payment had been requested. Separately, all respondents were asked how much they trusted property-tax officers to value properties fairly, with answers ranging from "not at all" to "completely," and how frequently officers requested or accepted informal payments during assessments in their area. These questions distinguish respondents' own experience from their general perceptions of tax administration.

Beliefs about satellite assessment. Three questions elicit expected accuracy, impartiality, and ease of correction. For accuracy, respondents considered ten houses in their neighbourhood similar to their own and reported how many a computer using satellite images would measure approximately correctly. For impartiality, respondents indicated their agreement with the statement that satellite valuation would treat their property like any other, without favouritism. For correction, they rated how easy or difficult it would be to have an erroneous computer assessment corrected. The latter two questions use four-point response scales. These questions were asked after the information script and before the choice experiment.

Ownership, documentation, and digital familiarity. Respondents reported whether the property was held solely in their name or under another ownership arrangement. They also reported whether they possessed and could produce within approximately one week the registry or sale deed, architectural plans recording land and covered area, and the latest property-tax receipt or PT-1. Digital familiarity was measured by respondents' reported comfort with completing an online form. The survey additionally elicited their likelihood of applying to check or correct the property's tax record within three months, on a scale from zero to ten. Three true–false items concerning courtesy, admission of mistakes, and resentment were included to account for differences in socially desirable responding.

Demand for property information. After completing the choice experiment, respondents were offered computer-estimated information about their own property, including covered area and the plot outline. The offered price was Rs 0, Rs 100, or Rs 200. Respondents who accepted paid the stated price through a deduction from their Rs 2,000 participation payment. This decision measures take-up of independent property information at the offered price. Results are reported in Appendix E.1.

5 Empirical Strategy

The analysis connects recorded choices to conditional attribute valuations, differences across taxpayers and the allocation of hypothetical tax claims. The primary panel mixed logit supplies the probability contrasts, monetary valuations, fiscal calculations and characteristic heterogeneity in Sections 6.1–6.4. The reported primary results use the same fitted specification.

The study comprises 2,100 respondents and 21,000 paired tasks: four fixed and six adaptive tasks per respondent. Task-specific eligibility determines the descriptive fixed-screen comparisons in Section 6.1; all valid choices enter the primary likelihood. For owners flagged as exempt, Q_i is the hypothetical reference liability displayed in the experiment.

5.1 Estimating Taxpayer Choices

For owner i, task t presents the two offers in $\mathcal{J}_{it}$. The outcome y_{ijt} is one if offer j is selected and zero otherwise. Let S_{ijt} indicate satellite assessment and D_{ijt} a positive informal payment. Relative to reference bill Q_i, define initial overassessment by $O_{ijt} = \mathbf{1}\{I_{ijt} > Q_i\}$ and underassessment by $U_{ijt} = \mathbf{1}\{I_{ijt} < Q_i\}$. Total expenditure is $P_{ijt} = T_{ijt} + B_{ijt}$, measured in rupees; the expenditure regressor is $P_{ijt}/1000$; Pos_{ijt} equals one for the second displayed receipt. I use the following index to relate the displayed attributes to choice:

$$\begin{aligned} v_{ijt} = {} & \alpha_i S_{ijt} + b_p P_{ijt}/1000 + \kappa_i D_{ijt} + \gamma_O O_{ijt} + \gamma_U U_{ijt} \\ & + \delta_O S_{ijt} O_{ijt} + \delta_U S_{ijt} U_{ijt} + \lambda_O D_{ijt} O_{ijt} + \lambda_U D_{ijt} U_{ijt} + \pi \text{Pos}_{ijt}. \end{aligned} \tag{10}$$

The reference offer is an inspector with a correct initial bill and no informal payment. A positive utility coefficient makes the stated attribute more attractive, holding expenditure and the other attributes fixed. The coefficients are in utility units; Sections 5.2 and 6 translate them into rupees and choice probabilities.

At a correct bill, α_i is the satellite premium over this reference and κ_i is the premium for an inspector offering a concession in exchange for an informal payment. Both vary across taxpayers and remain constant across a taxpayer's tasks. The expenditure coefficient b_p is the utility change from paying an additional Rs 1,000; a negative value means that higher payment reduces attractiveness. The position coefficient π is the utility change from displaying an otherwise unchanged offer second.

For an initial error $e \in \{O, U\}$, γ_e measures its effect on an inspector offer without a concession. The interaction δ_e measures how that error changes the satellite premium, and λ_e how it changes the concession premium. Thus the total error effect is $\gamma_e + \delta_e$ for satellite and $\gamma_e + \lambda_e$ for an inspector concession. A positive λ_e means that the concession becomes more attractive relative to an inspector without one when that initial error is present; the value of receiving a correct bill instead depends on the full sum $\gamma_e + \lambda_e$.

Conditional on a taxpayer's preferences, the probability of choosing one receipt is its exponentiated utility divided by the sum for the pair. Reported population predictions average these probabilities over the fitted distribution of preferences and across respondents. Order-balanced predictions also average the two display orders.

The BACE design utility selected candidate choice pairs using different parameters and units. Appendix C documents that utility separately from the analysis specification above, following the adaptive-design distinction in Drake et al. (2025b).

The conceptual restrictions in Appendix A translate into testable coefficient restrictions. C1, savings alone, implies $\bar{\kappa} = \lambda_O = \lambda_U = 0$ for mean concession valuation; savings alone for every taxpayer would additionally require zero characteristic shifts and residual heterogeneity in κ_i. C2, concession valuation unchanged across correct, excessive and understated initial bills, implies $\lambda_O = \lambda_U = 0$. C3, lower required saving following overassessment, predicts $\lambda_O > 0$, and C4, greater relative inspector valuation following overassessment, predicts $\delta_O < 0$. Joint Wald tests assess C1 and C2. Two-sided coefficient tests and intervals assess C3 and C4 against zero; their signs address the directional predictions. A direct test of $\lambda_O - \lambda_U = 0$ compares how excessive and understated initial bills change the concession premium. All tests use the common fit and its full respondent-level covariance.

Details of the panel likelihood, assumptions for the adaptive choice sequence, statistical inference and construction of predicted choice probabilities are provided in Appendix C.1.

5.2 Monetary Valuations and Fiscal Accounting

Converting utility coefficients into rupees. Let b collect the primary model's mean and characteristic-interaction coefficients and let $a_i'b$ denote a specified mean utility contrast for respondent i, excluding expenditure and receipt position. Its monetary valuation is the additional expenditure that offsets the attribute advantage under the common linear expenditure slope. With $\lambda_C = -b_p/1000$ denoting the marginal value of a rupee, the indifference condition and conversion are

$$a_i'b + \frac{b_p}{1000} WTP_i = 0, \qquad WTP_i = -1000\frac{a_i'b}{b_p}, \qquad \overline{WTP} = \frac{1}{N}\sum_{i=1}^{N} WTP_i. \tag{11}$$

The factor of 1,000 converts the model's expenditure units to rupees. A positive valuation is the extra expenditure that offsets the attribute benefit; a negative valuation is the expenditure

reduction required to offset the attribute disadvantage. These monetary measures express the fitted preference coefficients on an economically interpretable scale, as in the valuation exercises of Maestas et al. (2023a) and Wiswall and Zafar (2017); equation (11) applies the linear expenditure specification used here.

Valuations for the economic comparisons. For satellite versus inspector assessment without a concession, the sample-average numerator is $\bar{\alpha}$ at a correct bill and $\bar{\alpha} + \delta_e$ with initial error e. For inspector offers with versus without informal-payment concessions it is $\bar{\kappa}$ and $\bar{\kappa} + \lambda_e$, respectively. Initial-error contrasts use γ_e for an inspector without an informal payment, $\gamma_e + \lambda_e$ for an inspector offering a concession for an informal payment and $\gamma_e + \delta_e$ for the satellite. A correct-bill-versus-error comparison reverses the sign of the corresponding initial-error contrast. Thus the correct-minus-error valuation among inspector concessions is $1000(\gamma_e + \lambda_e)/b_p$, and its counterpart without a concession is $1000\gamma_e/b_p$. For a characteristic change Δx_r, the numerator is $\Delta x_r \eta_{Sr}$ or $\Delta x_r \eta_{Dr}$.

Uncertainty in monetary valuations. Uncertainty uses the full covariance of the numerator and b_p. With $\bar{a} = N^{-1} \sum_i a_i$ and $g(b) = -1000\bar{a}'b/b_p$, the delta-method variance is $\nabla g(b)' \widehat{\Sigma} \nabla g(b)$. Fieller intervals use the joint covariance of $\bar{a}'b$ and b_p to form the ratio interval. Rupee valuations average utility-to-expenditure ratios; choice shares average the nonlinear choice probabilities. The two summaries can differ in their ranking around zero and one-half when preferences vary across respondents.

Expected payments and taxpayer savings. For fiscal accounting, let $\bar{p}_i$ be the integrated, order-balanced probability of selecting the inspector concession from the same receipts used in the central choice comparisons. For component $X \in \{T, B, P\}$, expected payment is

$$E[X_i] = (1 - \bar{p}_i) X_{iS} + \bar{p}_i X_{iH}, \qquad E[P_i] = E[T_i] + E[B_i]. \tag{12}$$

Taxpayer saving relative to correct liability is $Q_i - E[P_i]$. The owner-normalised outcomes average $100E[X_i]/Q_i$ equally across respondents. Thus both choices and fiscal allocations use respondent-specific Q_i, with the same weights and receipt order convention.

Reference receipts and fiscal components. Section 6.4 follows Panel B's correct satellite against an inspector with a correct or excessive initial bill, at a 10% net saving. The inspector's receipt is $T_{iH} = 0.7Q_i$, $B_{iH} = 0.2Q_i$ and $P_{iH} = 0.9Q_i$. An excessive initial claim is $1.1Q_i$. Its reduction to $0.7Q_i$ removes $0.1Q_i$ of excess and then reduces official payment by $0.3Q_i$ below correct liability. Selection probabilities weight these components. The payment split is a scenario assumption; the model estimates which arrangement is selected. These calculations describe hypothetical private payments and official receipts.

5.3 Preference Heterogeneity and Institutional Beliefs

The primary model allows assessor and concession preferences to vary with observed characteristics and residual individual differences (Mas and Pallais, 2017b).

To examine how beliefs and circumstances relate to administrative valuations, the common panel mixed logit includes their interactions from the outset using the full choice sequence from all 2,100 respondents. The ten domains are expected correction ease, expected satellite accuracy and impartiality, officer trust, prior departmental contact, perceived informal payments, digital comfort, title status, document possession and reference-bill scale. Each characteristic interacts with the satellite and concession indicators. These interactions allow the mean assessor and concession coefficients to vary with measured characteristics; the initial-error coefficients remain common. Persistent correlated normal variation in assessor and concession preferences remains after accounting for these characteristics.

All ten domains enter jointly, together with the displayed offer attributes, expenditure, receipt position and the three-item social-desirability score interacted with satellite and concession indicators. Non-substantive characteristic responses are treated as missing, replaced by the observed respondent mean for that characteristic, and centred over respondents. This retains all valid choices in the common fit. The opening choices contribute to the likelihood alongside later choices; their answers are not predictors. The same sequential-design assumptions in Section C.1 apply. Expected correction, accuracy and impartiality are measured after the information script but before BACE; the estimates describe conditional associations.

For Table 6, I extend this choice model with paid-information acceptance interacted with the satellite and concession indicators. The same utility also includes free-offer acceptance, offered-price indicators and recorded information-treatment indicators interacted with both arrangements. This augmented fit uses the full recorded choice sequences of the 2,099 respondents with substantive offer responses. All rows of the table come from this joint fit.

The table reports changes in rupee valuation for the stated characteristic contrasts, at correct initial bills and with other included characteristics held fixed. Paid acceptance compares acceptors with decliners among the Rs 100 and Rs 200 offers, conditional on offered price. Characteristic shifts are common across initial-bill conditions, as specified in the utility. Appendix B.2 gives the characteristic coding. Holm adjustment covers the twenty-two displayed monetary contrasts; direct satellite-minus-concession comparisons use a separate family of eleven contrasts. Pointwise intervals and tests use the augmented fit's full respondent-score covariance.

6 Results

6.1 Taxpayers value a correct initial bill

The opening choices show how assessor identity and private savings enter the decision (Figure 1). At equal correct bills and equal payment, 59.8% choose satellite over an inspector. In a separate

comparison between two inspectors, 55.1% of survey respondents choose the offer that lowers total payment in exchange for an informal payment. The latter share measures acceptance of the inspector's bill-reduction arrangement against another inspector offer. Receipt order is fixed in these opening tasks, and each share uses its own displayed-receipt eligibility rule.

Taxpayers value getting the assessment right before any bill reduction, even when the eventual payment is held equal. Figure 1 shows this preference in comparisons between two inspectors who both reduce the official bill in exchange for an informal payment. With almost identical total payment, 76.9% choose the initially correct bill over an excessive bill followed by a reduction (N = 2,060; 95% interval 75.0–78.7%). When both total expenditure and informal payment are exactly equal, 74.0% choose the initially correct bill. These comparisons combine baseline values that survey respondents place on the initial accuracy and the concession offered after a correct, excessive or understated initial bill.

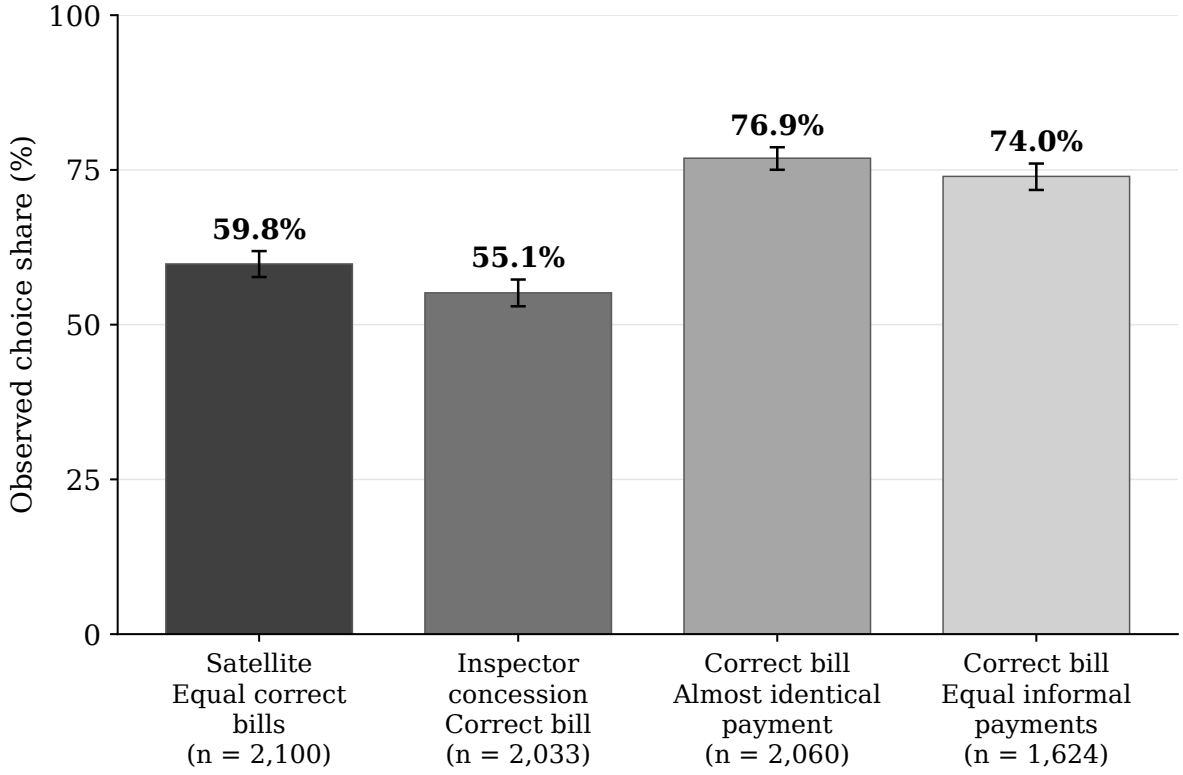


Figure 1: Choices in the four common scenarios

Notes: Observed shares and Wilson 95% intervals. Eligible denominators appear below each bar. Satellite is first in screen 1, the concession second in screen 2, and the correct bill first in screens 3–4. Screen 3 compares two inspector concessions with almost identical payment; screen 4 holds both expenditure and informal payment equal. These descriptive eligibility rules leave the full-model sample unchanged.

Reading the monetary valuations. Table 4 expresses the fitted preferences in rupees for the full choice set of 21000 from 2100 survey respondents. Each row values the first arrangement relative to the second, holding total expenditure equal. A positive entry is the extra valuse in monetary terms that taxpayers place on the first choice relative to the comparison choice;

a negative entry is the payment reduction needed to make it equally attractive. In the table, a *concession* means an official-bill reduction in exchange for an informal payment to the inspector. The first row compares a satellite with an inspector without offering any concession, at the same initial bill. The next three rows compare an officer offering a concession against an inspector offering any concession. The last two compare initial bills when both inspectors offer reductions for informal payments.

Table 4: Monetary valuations: assessment and concession comparisons (Rs)

Comparison	Valuation (Rs)	95% confidence interval	p
Satellite vs inspector without concession: correct bill	-184	[-486, 131]	0.227
Inspector with vs without concession: correct bill	65	[-303, 615]	0.770
Inspector with vs without concession: excessive bill	925**	[269, 1,854]	0.016
Inspector with vs without concession: too-low bill	866**	[274, 1,697]	0.012
Both inspectors offer concessions: Correct vs excessive initial bill	1,897***	[1,463, 2,596]	< 0.001
Both inspectors offer concessions: Correct vs too-low initial bill	503**	[126, 961]	0.014

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. Entries value the first arrangement relative to the second, holding total expenditure equal. Positive values favour the first arrangement; negative values favour the second. Initial bills are held equal in the first four rows; both inspectors offer concessions in the last two. Common full-sequence panel mixed logit; 2,100 respondents. The study design comprises 21,000 paired tasks: four fixed and six adaptive tasks per respondent. Inference uses the full respondent-score covariance. Brackets give pointwise Fieller 95% confidence intervals. Tests use the full covariance and are two-sided; p-values are unadjusted. Stars indicate * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The average satellite premium in the primary model. At a correct bill, the mean satellite valuation relative to an inspector without a bill-reduction arrangement is Rs -184 [-486, 131]. The coefficient is statistically insignificant. This means that after initially preferring satellite, survey respondents don't place extra value on technology-based assessment of their tax liabilities after going through all 10 sets of choices. The corresponding model-predicted satellite share at identical receipts and balanced order is 50.4% (Appendix E.2). The opening share records observed choices in their displayed order; this fitted share averages the two receipt orders.

Bill reductions are valued differently after correct and incorrect assessments. Relative to an inspector without a bill-reduction arrangement, the estimated valuation of an inspector offering an informal payment for a correct initial bill is Rs 65. However, the estimate is statistically insignificant ($p - value = 0.770$). This means that if the tax liability is correct, taxpayers will not prefer to pay an informal payment to the tax officer. When the initial bill is excessive, taxpayers are willing to pay an additional Rs 925 for an inspector who offers to reduce the official bill in exchange for an informal payment, relative to an inspector who offers no reduction. This

valuation is positive and statistically significant at the 5% level ($p = 0.016$). Holding the initial bill and total expenditure equal, the estimate indicates a preference for access to a negotiated reduction beyond its cash saving. Here, reducing the official bill can remove part or all of the excess demand.

When the initial bill is already below the correct liability, taxpayers are willing to pay an additional Rs 866 for an inspector offering a further reduction in exchange for an informal payment, relative to an inspector offering no reduction. This valuation is also positive and statistically significant at the 5% level ($p = 0.012$). The initial underassessment precedes the negotiated reduction. In this case, the positive valuation concerns an offer that takes the official payment further below the correct liability, extending the taxpayer's private benefit while deepening undercollection. This comparison also holds the initial bill and total expenditure equal. The joint test rejects equal mean valuations across the three initial-bill conditions (Appendix E.2).

Accuracy has value beyond the final payment. When both inspectors offer to reduce the official bill in exchange for an informal payment, taxpayers prefer the inspector who starts with the correct assessment.

Compared with an excessive initial bill, taxpayers are willing to pay Rs 1,897 more for a correct initial bill, holding final total expenditure equal. This valuation is statistically significant at the 1% level (95% confidence interval: Rs 1,463–2,596). Reaching the same final payment through a negotiated reduction therefore does not remove the preference for being assessed correctly from the outset.

Compared with an initial bill below the correct liability, taxpayers are willing to pay Rs 503 more for a correct initial bill, again holding final total expenditure equal. This valuation is statistically significant at the 5% level (95% confidence interval: Rs 126–961). Here, the lower initial bill exists before either inspector offers a reduction. The positive valuation indicates that taxpayers prefer a correct starting assessment even to an understated one when both offers involve informal payments and lead to the same total expenditure. Thus, the preference for initial accuracy extends to errors in both directions.

The two findings concern different choices. At a given excessive bill, the taxpayer can value the inspector's offer of a reduction relative to an inspector without that offer. When both inspectors offer reductions, the taxpayer can still prefer the one who starts with the correct bill. The first comparison values the arrangement conditional on the initial assessment; the second values initial accuracy among those arrangements.

6.2 Accuracy competes with savings from the inspector

Taxpayers value an accurate initial assessment, but their choice also depends on what they can save through the inspector. Table 5 brings these two considerations together. Every cell in Panels A–B reports the model-predicted percentage choosing satellite over an inspector who

offers to reduce the official bill in exchange for an informal payment. The remaining percentage chooses the inspector. A higher number therefore means greater take-up of satellite assessment.

Read down the rows to increase the taxpayer's saving from choosing the inspector: 5%, 10% or 20% of correct liability. This saving is calculated after paying the informal payment. The informal payment stays fixed at 20% of correct liability; larger savings come from a lower official payment. Panel A gives both assessors the same initial assessment. Panel B keeps satellite correct and shows excessive or understated inspector assessments. The case in which both assess correctly appears once, in Panel A, column (1), and provides the common benchmark for both panels. These are bills before the inspector offers any reduction. Predictions average over respondents and their preference distributions, with both receipt orders equally weighted. Panel C reports changes in satellite choice in percentage points.

Panel A: private savings when both assessors start with the same assessment. Column (1) holds both initial bills correct. Reading down this column, predicted satellite choice falls from 44.2% to 39.6% to 32.7% as the taxpayer's saving from choosing the inspector rises from 5% to 10% to 20%. Larger financial benefits make the inspector more attractive even when satellite assesses correctly. Columns (2) and (3) repeat the comparison when both initially assess 10% too high or 10% too low. At a 10% saving, satellite choice is 34.6% in column (2) and 37.5% in column (3), compared with 39.6% in column (1). In each column, both assessors share the initial assessment; satellite has no initial-accuracy advantage over the inspector.

Panel B: accurate satellite assessment against an incorrect inspector assessment. Column (4) keeps satellite correct and the inspector's initial bill 10% excessive. Predicted satellite choice is 54.0% at a 5% inspector saving, 49.1% at 10%, and 41.3% at 20%. Accurate satellite assessment attracts a predicted majority at the smallest saving, but larger savings draw choices towards the inspector. At a 10% saving, comparing column (4) with the both-correct benchmark in Panel A, column (1), raises satellite choice from 39.6% to 49.1%. The final payment terms are identical; only the inspector's initial assessment changes. Column (5) instead gives the inspector an initial bill 10% below correct liability. At the reported 20% saving, 35.0% choose satellite and the remainder choose the inspector.

Panel C: separating the savings and accuracy comparisons. Column (6) reports the changes in satellite choice, with 95% intervals in column (7). Row C1 compares the 20% and 5% savings rows of Panel A, column (1): the larger saving reduces satellite choice by 11.4 percentage points. Row C2 compares columns (2) and (1) at a 10% saving: making both initial assessments excessive reduces satellite choice by 5.1 percentage points. Row C3 compares Panel B, column (4), with Panel A, column (1): making only the inspector's initial assessment excessive raises satellite choice by 9.5 percentage points. All three differences are statistically significant at the 1% level.

Table 5: Predicted probability of choosing satellite rather than an inspector offering a bill reduction (%)

A. Both assessors start with the same initial assessment

	Outcome variable: Pr(choosing satellite), %		
Taxpayer's saving from choosing inspector rather than satellite (% of correct liability)	(1) Both initially assess correctly	(2) Both initially assess 10% too high	(3) Both initially assess 10% too low
5% saving	44.2*** [42.3, 46.1]	38.8*** [37.1, 40.5]	42.0*** [40.2, 43.8]
10% saving	39.6*** [37.7, 41.6]	34.6*** [32.9, 36.2]	37.5*** [35.6, 39.5]
20% saving	32.7*** [30.4, 35.1]	28.3*** [26.3, 30.2]	–

B. Satellite assesses correctly; inspector initially assesses incorrectly

	Outcome variable: Pr(choosing satellite), %	
Taxpayer's saving from choosing inspector rather than satellite (% of correct liability)	(4) Inspector initially assesses 10% too high	(5) Inspector initially assesses 10% too low
5% saving	54.0*** [52.0, 56.1]	–
10% saving	49.1 [46.8, 51.4]	–
20% saving	41.3*** [38.4, 44.2]	35.0*** [32.0, 37.9]

C. Changes in Pr(choosing satellite), percentage points

Change being compared	(6) Change (percentage points)	(7) 95% interval
C1. Increase saving from 5% to 20%; both assess correctly (column 1)	-11.4***	[-13.3, -9.6]
C2. Both initial assessments: correct → excessive; 10% saving (column 2 minus column 1)	-5.1***	[-7.1, -3.1]
C3. Inspector initial assessment: correct → excessive; satellite correct; 10% saving (column 4 minus column 1)	9.5***	[8.4, 10.5]
C4. Satellite initial assessment: excessive → correct; inspector excessive; 10% saving (column 4 minus column 2)	14.5***	[12.3, 16.8]

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. Panels A–B give the predicted percentage choosing satellite over the inspector concession. Saving is satellite payment minus the inspector's total payment, as a percentage of correct liability. Panel C reports the change in satellite choice for each stated comparison, in percentage points. Two-sided tests are unadjusted: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Row C4 isolates the improvement in satellite's initial assessment while the inspector remains excessive. Comparing Panel B, column (4), with Panel A, column (2), at a 10% saving raises satellite choice from 34.6% to 49.1%. The increase is 14.5 percentage points (95% interval: 12.3–16.8), statistically significant at the 1% level. Both total payments fall equally between these scenarios, preserving the inspector's saving and informal payment. Under the common linear expenditure slope, the model attributes the increase to satellite's improved initial accuracy. Almost half of taxpayers are now predicted to choose satellite despite the inspector's lower total payment.

Extending the saving comparison in Panel B, column (4), accurate satellite assessment reaches a 50% predicted share when the inspector saving is 9.0% of correct liability (95% confidence set: 6.9–11.7%). This threshold describes an even split in the average predicted choice, not an average individual reservation saving. With expenditure measured relative to liability, it is 8.0% [7.2, 8.8]. Accurate technology attracts taxpayers facing excessive inspector demands, while sufficiently large negotiated savings draw them towards the inspector. The next section translates these choices into official receipts, informal payments and taxpayer savings.

6.3 How monetary valuations differ across taxpayers

The average valuations coexist with differences across taxpayers. Table 6 relates these differences to institutional expectations, taxpayer circumstances and paid demand for property information. Read each row from left to right: reference → comparison. Each entry is the valuation at the arrow's endpoint minus the valuation at its starting point, holding other characteristics fixed. Both columns value the named arrangement relative to an inspector without this bill-reduction arrangement, at correct initial bills and equal total expenditure. Positive entries mean higher valuation for the comparison group; negative entries mean lower valuation. These beliefs and circumstances are not randomised mechanisms.

Table 6: Differences in monetary valuation across taxpayers (Rs)

Characteristic: reference → comparison	Change in satellite valuation (Rs)	Change in inspector concession valuation (Rs)
A. Institutional expectations		
Officer trust: a little → mostly	-954***	251
	[-1,474, -545]	[-92, 618]
Correcting computer errors: very hard → easy	2,120***	1,757***
	[1,275, 3,201]	[1,020, 2,682]
Expected satellite accuracy: 4 → 6 of 10 houses	558**	-417*
	[207, 966]	[-783, -106]
Satellite fairness: disagree → agree	4,140***	252
	[3,275, 5,482]	[-181, 721]
Informal-payment prevalence: rarely → often	1,115***	2,250***
	[504, 1,836]	[1,631, 3,105]
B. Taxpayer circumstances, documentation and bill size		
Administrative contact last year: no → yes	2,930***	1,934***
	[2,154, 4,066]	[1,396, 2,629]
Online-form comfort: a little → comfortable	-651***	896***
	[-1,042, -328]	[613, 1,260]
Property title: sole → non-sole	-1,337***	11
	[-2,118, -715]	[-532, 561]
Documentation score: $1/3 \rightarrow 1$	-856*	-1,519***
	[-1,627, -216]	[-2,235, -943]
Reference liability: Rs 5,000 → Rs 25,000	-1,368***	-1,513***
	[-1,864, -949]	[-1,933, -1,118]
C. Paid demand for property information		
Paid information offer: declined → accepted	1,120***	639
	[495, 1,841]	[81, 1,257]

Notes: Each row reads reference → comparison; entries are comparison-minus-reference differences in rupee valuation, holding other characteristics fixed. Positive values indicate higher valuation at the arrow's endpoint; negative values indicate lower valuation. Each valuation compares with an inspector without a concession, at correct initial bills and equal expenditure. A concession reduces the official bill in exchange for an informal payment. All rows come from one augmented panel mixed logit (2,099 respondents). Controls also include social desirability, free-offer acceptance, offered price and information treatment, interacted with both arrangements. Paid acceptance compares 561 acceptors with 870 decliners at Rs 100 or Rs 200, conditional on price. Brackets: pointwise Fieller 95% intervals. Stars: full-covariance two-sided tests, Holm-adjusted across twenty-two contrasts; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Non-sole title includes joint, family, disputed and no-record categories. Documentation averages valid registry, plan and tax-receipt possession indicators. Missing characteristics retain mean imputation.

Fairness expectations distinguish satellite assessment. Agreeing rather than disagreeing that satellite assessment treats properties fairly is associated with a Rs 4,140 higher satellite valuation. The corresponding association with the inspector's bill-reduction arrangement is small and statistically insignificant after adjustment. The directly estimated difference between the two associations is Rs 3,888 [2,919, 4,857]. Fairness expectations therefore relate particularly strongly to the valuation of satellite assessment relative to the inspector alternative.

Expected correction relates to both arrangements. Respondents who expect correction to be easy rather than very hard have a satellite valuation Rs 2,120 higher and a valuation of the inspector arrangement Rs 1,757 higher. The directly estimated difference is Rs 363 [-406, 1,132], with a Holm-adjusted p-value of 0.710. Easier expected correction thus relates positively to both alternatives to the reference inspector, while the estimated difference between those associations is imprecise.

Administrative experience and taxpayer circumstances. Greater trust in officers is associated with lower satellite valuation, while prior departmental contact is associated with higher valuations of both arrangements. Expected satellite accuracy is positively associated with satellite valuation and negatively associated with valuation of the inspector arrangement. Digital comfort has the reverse pattern: lower satellite valuation and higher valuation of the inspector arrangement. Non-sole title is associated with lower satellite valuation; the documentation and reference-liability estimates are negative in both columns. Digital comfort, documentation and title have different patterns; their associations do not support a single administrative-capacity interpretation.

6.3.1 Paid information demand and satellite preferences

Taxpayers who accepted the paid property-information offer also place a higher value on satellite assessment. Among respondents offered information for Rs 100 or Rs 200, acceptors have a satellite valuation Rs 1,120 higher than decliners (95% interval: Rs 495–1,841; Table 6). The comparison values satellite relative to an inspector without an informal payment, with both initial bills correct and total expenditure equal. It holds the other included characteristics, offered price and information treatment fixed, and survives the table's multiple-testing adjustment.

Accepting the offer involved a deduction from the survey participation payment. The result therefore links a financially consequential decision to obtain independent property information with higher choice-based valuation of satellite assessment.

Together, the results connect demand for a correct initial assessment to the private savings available through inspectors. Accuracy and financial terms shape predicted choices; those choices determine expected official and informal payments under the specified receipts. Fairness, correction expectations and paid information demand describe important differences in valuations across taxpayers. The institutional implication is to provide accurate assessment and legitimate correction without requiring private bargaining. That is a policy implication of the preference evidence, rather than an effect of a reform measured here.

6.4 Introducing accurate technology: official revenue and private savings

Introducing accurate satellite assessment alongside inspectors gives taxpayers a choice between paying the correct liability officially and obtaining a negotiated bill reduction in exchange for an

informal payment. Table 7 shows how predicted choices divide payments between the administration and inspectors, and how much taxpayers retain. The policy scenario gives taxpayers access to both options, with satellite always assessing correctly.

Each entry is an expected amount per Rs 100 of correct liability, averaging the two offers using their predicted selection probabilities. Satellite collects Rs 100 officially and no informal payment. Choosing the inspector instead saves the taxpayer Rs 5, Rs 10 or Rs 20 across Panels A–C, after an informal payment of Rs 20. The inspector's corresponding official payments are Rs 75, Rs 70 and Rs 60. Within each panel, these final payment terms stay fixed while the inspector's initial bill changes from correct to 10% excessive. Comparisons between the two rows therefore show how the initial assessment changes predicted selection and the resulting allocation of payments.

Table 7: Predicted payment allocations when accurate satellite assessment and inspectors are both available

Expected amounts per Rs 100 of correct liability				
Inspector's initial assessment	Official revenue	Informal payment	Total taxpayer payment	Taxpayer saving
A. Taxpayer saves 5% of correct liability by choosing inspector				
Correct initial bill	86.05	11.16	97.21	2.79
	(85.57, 86.52)	(10.78, 11.54)	(97.11, 97.30)	(2.70, 2.89)
Initial bill 10% too high	88.51	9.20	97.70	2.30
	(87.99, 89.02)	(8.78, 9.61)	(97.60, 97.80)	(2.20, 2.40)
B. Taxpayer saves 10% of correct liability by choosing inspector				
Correct initial bill	81.89	12.07	93.96	6.04
	(81.31, 82.47)	(11.68, 12.46)	(93.77, 94.16)	(5.84, 6.23)
Initial bill 10% too high	84.73	10.18	94.91	5.09
	(84.04, 85.43)	(9.71, 10.64)	(94.68, 95.14)	(4.86, 5.32)
C. Taxpayer saves 20% of correct liability by choosing inspector				
Correct initial bill	73.10	13.45	86.55	13.45
	(72.17, 74.02)	(12.99, 13.92)	(86.08, 87.01)	(12.99, 13.92)
Initial bill 10% too high	76.53	11.74	88.26	11.74
	(75.36, 77.70)	(11.15, 12.32)	(87.68, 88.85)	(11.15, 12.32)

Notes: All amounts are Rs per Rs 100 of correct liability. Each cell averages payments or savings across predicted satellite and inspector choices. Satellite always assesses correctly and collects Rs 100 officially, with no informal payment. In Panels A, B and C, choosing the inspector yields net taxpayer savings of Rs 5, Rs 10 and Rs 20, respectively: official payments are Rs 75, Rs 70 and Rs 60, and the informal payment is Rs 20 throughout. Within each panel, only the inspector's initial assessment changes. Predictions use the Panel B offers in Table 5, equal respondent weights and balanced order. Official revenue plus informal payment equals total taxpayer payment; taxpayer saving equals Rs 100 minus total payment. Common full-sequence panel mixed logit; 2,100 respondents. The study design comprises 21,000 paired tasks: four fixed and six adaptive tasks per respondent. Inference uses the full respondent-score covariance. Parentheses give pointwise 95% delta-method intervals.

Private savings draw payments away from the administration. Across the three panels, larger savings from choosing the inspector reduce expected official revenue and increase expected informal payments. Two forces operate together: the inspector collects less official tax per transaction, and more taxpayers choose the inspector over satellite. Accurate assessment therefore competes with the financial benefits taxpayers can obtain through negotiated reductions. The informal payment per inspector transaction remains fixed; its expected amount rises because the inspector is selected more often.

Excessive inspector demands increase the appeal of accurate technology. At each saving level, an excessive initial inspector bill shifts predicted choices towards satellite. In the 10% saving panel, the inspector's predicted share falls from 60.4% to 50.9%. Expected official revenue consequently rises from Rs 81.89 to Rs 84.73 per Rs 100 of liability, while expected informal payment falls. The higher official receipts arise because more taxpayers choose accurate satellite assessment, which collects the full liability officially. The inspector's final official payment and informal payment are unchanged between these two rows.

The policy implication is that the appeal of accurate technology depends on both the initial assessments taxpayers face and the private savings available through inspectors. An accurate alternative attracts more predicted take-up when inspectors initially overassess, directing a larger share of payments to the administration. Larger negotiated savings pull choices in the opposite direction. In this proposed choice system, these competing preferences determine expected official receipts and informal payments.

An excessive inspector demand also gives the subsequent official-bill reduction two components: removing the excess demand and reducing payment below the correct liability. In the 10% saving panel, their expected amounts are Rs 5.09 and Rs 15.27 per Rs 100, respectively, after weighting by the predicted inspector share.

7 Discussion and Conclusion

Digital tax administration is increasingly central to efforts to strengthen revenue collection in developing countries. By improving the information available to tax authorities and standardising administrative decisions, digital systems can help establish liabilities more accurately and limit discretionary departures from tax rules. Yet their introduction also changes the arrangements through which taxpayers obtain assistance, contest demands and negotiate payments. This paper examines taxpayers' preferences over these arrangements and the private incentives that may influence the acceptability of administrative reform.

The findings distinguish preferences over the identity of the assessor from the value of accurate assessment and the attraction of discretionary settlement. Across the full choice experiment, the average valuation of satellite relative to inspector assessment is statistically insignificant. Taxpayers nevertheless value receiving a correct initial bill, even when final expenditure

is unchanged. They also respond to the financial savings offered through inspector concessions, while their valuation of those arrangements varies with the initial assessment state. These results suggest that the appeal of technology depends on the administrative outcomes it delivers and the alternatives available through officials.

The comparisons between complete arrangements make this trade-off concrete. When an inspector offers a net saving of 10 percent of reference liability, providing a correct satellite bill rather than an excessive one raises predicted satellite selection by 14.5 percentage points under the specified comparisons. Against an initially excessive inspector bill followed by a concession, the population-average probability of choosing accurate satellite assessment reaches one-half at an inspector saving of approximately 9 percent. Accuracy therefore gives technology an advantage that private financial concessions can offset. This advantage does not require a positive average preference for the satellite assessor itself.

The distinction matters because the benefits available through frontline officials have different implications for public finance. Correcting an excessive bill brings the administrative claim into line with the amount owed. Reducing official payment below the correct liability instead diverts part of that claim into taxpayer savings and informal payments. The fiscal calculations show how predicted choices between the specified arrangements translate into these different allocations. Their magnitudes depend on the offered terms and do not measure actual revenue losses from discretion. They nevertheless clarify why an arrangement that is privately attractive to taxpayers can reduce official receipts by more than the saving taxpayers retain.

Institutional expectations help place these choices within the taxpayer–administration relationship. Perceived fairness of satellite assessment and expected access to correction are positively associated with satellite valuation, while greater trust in tax officers is associated with lower relative satellite valuation. Expected correction is also positively associated with concession valuation. Confidence in obtaining a satisfactory administrative outcome can therefore accompany attraction to either arrangement. These associations do not establish the effects of improving trust or correction procedures, but they indicate why preferences over technology cannot be understood independently of taxpayers' expectations about the institution responsible for their bills.

The findings inform the changing role of frontline officials as tax administrations move from street-level toward system-level decision-making. Technology can assume a greater role in establishing taxable facts and determining liabilities, while officials remain responsible for explaining decisions, reviewing evidence and resolving disputes. The institutional challenge is to make legitimate correction accessible while restricting privately negotiated departures from the tax claim. Accurate assessment and credible recourse may strengthen the acceptability of digital administration, but they need not eliminate taxpayers' financial interest in concessions. Reform must therefore address both the quality of administrative decisions and the authority governing how those decisions become payment obligations.

More broadly, the paper places taxpayers' preferences and private incentives within the political economy of fiscal capacity. Improvements in information and assessment technology operate through relationships between revenue authorities, frontline officials and taxpayers. Those relationships can provide both legitimate assistance and opportunities to reduce payments below what is owed. The findings identify a demand-side constraint on reform: taxpayers can value accurate and fair administration while retaining an economic interest in discretionary settlement. Strengthening revenue collection consequently requires institutions that make correct treatment credible and accessible while limiting the private bargains through which public claims are reduced.

References

Autor, David H, Frank Levy, and Richard J Murnane, "The skill content of recent technological change: An empirical exploration," *The Quarterly journal of economics*, 2003, *118* (4), 1279–1333.

Ballard-Rosa, Cameron, Lucy Martin, and Kenneth Scheve, "The Structure of American Income Tax Policy Preferences," *The Journal of Politics*, 2017, *79* (1), 1–16.

Banerjee, Ritwik, "On the Interpretation of Bribery in a Laboratory Corruption Game: Moral Frames and Social Norms," *Experimental Economics*, 2016, *19* (1), 240–267.

Barr, Abigail and Danila Serra, "The Effects of Externalities and Framing on Bribery in a Petty Corruption Experiment," *Experimental Economics*, 2009, *12* (4), 488–503.

Besley, Timothy and Torsten Persson, "Why Do Developing Countries Tax So Little?," *Journal of Economic Perspectives*, 2014, *28* (4), 99–120.

Bolton, Gary E., Jordi Brandts, and Axel Ockenfels, "Fair Procedures: Evidence from Games Involving Lotteries," *The Economic Journal*, 2005, *115* (506), 1054–1076.

Bovens, Mark and Stavros Zouridis, "From street-level to system-level bureaucracies: How information and communication technology is transforming administrative discretion and constitutional control," *Public administration review*, 2002, *62* (2), 174–184.

Bullock, Justin B, "Artificial intelligence, discretion, and bureaucracy," *The American Review of Public Administration*, 2019, *49* (7), 751–761.

Bureau of Statistics, Government of the Punjab and UNICEF Punjab, "Multiple Indicator Cluster Survey, Punjab 2024: Survey Findings Report, Volume II: Districts Outlook," Technical Report, Bureau of Statistics, Planning and Development Board, Government of the Punjab and UNICEF Punjab, Lahore, Pakistan 2025. Table D SR.2.2, p. 21. Survey year: 2024.

Bureau of Statistics Punjab, "Multiple Indicator Cluster Survey Punjab 2011: Provincial Report, Volume I," Technical Report, Planning & Development Department, Government of the Punjab, Lahore, Pakistan 2024.

Carrillo, Paul, Dina Pomeranz, and Monica Singhal, "Dodging the Taxman: Firm Misreporting and Limits to Tax Enforcement," *American Economic Journal: Applied Economics*, 2017, *9* (2), 144–164.

Czajka, Léo, Awa Diouf, and Fabrizio Santoro, "Do Electronic Filing and Payment Increase Tax Compliance? Evidence from Large Taxpayers in Senegal," ICTD Working Paper 238, Institute of Development Studies 2026.

Drake, Marshall, Fernando Payró, Neil Thakral, and Linh T. Tô, "Bayesian Adaptive Choice Experiments," February 2025. Working paper.

_ , Fernando Payró, Neil Thakral, and Linh T. Tô, "Bayesian Adaptive Choice Experiments," 2025. Working paper, February.

_ , Neil Thakral, and Linh T. Tô, "Wage Differentials and the Price of Workplace Flexibility," 2025. Working paper, January.

Dzansi, James, Anders Jensen, David Lagakos, and Henry Telli, "Technology and Tax Capacity: Evidence from Local Governments in Ghana," Working Paper 29923, National Bureau of Economic Research 2022. Revised February 2025.

Frey, Bruno S., Matthias Benz, and Alois Stutzer, "Introducing Procedural Utility: Not Only What, but Also How Matters," *Journal of Institutional and Theoretical Economics*, 2004, *160* (3), 377–401.

Gangl, Katharina, Eva Hofmann, and Erich Kirchler, "Tax Authorities' Interaction with Taxpayers: A Conception of Compliance in Social Dilemmas by Power and Trust," *New Ideas in Psychology*, 2015, *37*, 13–23.

Government of Punjab, "The Punjab Finance Act 2024," Excise, Taxation and Narcotics Control Department, Government of Punjab 2024. Punjab Act XIII of 2024.

Gräfe, Philipp, Justine Marienfeldt, Liz Marla Wehmeier, and Sabine Kuhlmann, "Changing tasks and changing public servants? The digitalisation and automation of public administrative work," *Information Polity*, 2024, p. 13837605241289773.

Hassan, Mirza and Wilson Prichard, "The Political Economy of Tax Reform in Bangladesh: Political Settlements, Informal Institutions and the Negotiation of Reform," ICTD Working Paper 14, Institute of Development Studies 2013.

Khan, Adnan Q., Asim Ijaz Khwaja, and Benjamin A. Olken, "Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors," *The Quarterly Journal of Economics*, 2016, *131* (1), 219–271.

_ , _ , and _ , "Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors," *Quarterly Journal of Economics*, 2016, *131* (1), 219–271.

Kirchler, Erich, Erik Hoelzl, and Ingrid Wahl, "Enforced versus Voluntary Tax Compliance: The "Slippery Slope" Framework," *Journal of Economic Psychology*, 2008, *29* (2), 210–225.

Knebelmann, Justine, Victor Pouliquen, and Bassirou Sarr, "Discretion versus Algorithms: Bureaucrats, Tax Equity and Acceptability," Working Paper 26/34, Institute for Fiscal Studies 2026.

Luttmer, Erzo F. P. and Monica Singhal, "Tax Morale," *Journal of Economic Perspectives*, 2014, *28* (4), 149–168.

Maestas, Nicole, Kathleen J. Mullen, David Powell, Till von Wachter, and Jeffrey B. Wenger, "The Value of Working Conditions in the United States and Implications for the Structure of Wages," *American Economic Review*, 2023, *113* (7), 2007–2047.

_ , _ , _ , _ , and _ , "The Value of Working Conditions in the United States and the Implications for the Structure of Wages," *American Economic Review*, 2023, *113* (7), 2007–2047.

Mas, Alexandre and Amanda Pallais, "Valuing Alternative Work Arrangements," *American Economic Review*, 2017, *107* (12), 3722–3759.

_ and _ , "Valuing Alternative Work Arrangements," 2017. Author manuscript, March.

Mascagni, Giulia, Celeste Scarpini, Denis Mukama, Fabrizio Santoro, and Naphtal Hakizimana, "The Interpersonal Side of Tax Compliance: Interactions Between Taxpayers and Tax Officials in Rwanda," ICTD Working Paper 215, Institute of Development Studies 2025.

Murphy, Kristina, "The Role of Trust in Nurturing Compliance: A Study of Accused Tax Avoiders," *Law and Human Behavior*, 2004, *28* (2), 187–209.

Okunogbe, Oyebola and Gabriel Tourek, "How Can Lower-Income Countries Collect More Taxes? The Role of Technology, Tax Agents, and Politics," *Journal of Economic Perspectives*, 2024, *38* (1), 81–106.

__ **and Victor Pouliquen**, "Technology, Taxation, and Corruption: Evidence from the Introduction of Electronic Tax Filing," *American Economic Journal: Economic Policy*, 2022, *14* (1), 341–372.

Piracha, Mujtaba and Mick Moore, "Understanding Low-Level State Capacity: Property Tax Collection in Pakistan," ICTD Working Paper 33, International Centre for Tax and Development, Brighton 2015.

__ **and** __ , "Revenue-Maximising or Revenue-Sacrificing Government? Property Tax in Pakistan," *Journal of Development Studies*, 2016, *52* (12), 1776–1790.

Prichard, Wilson and Paul Fish, "Strengthening IT Systems for Property Tax Reform," ICTD Summary Brief 11, Institute of Development Studies 2017.

Santoro, Fabrizio, Phindile T. Masuku, and Tanele Magongo, "Cashless Tax Systems: Voluntary vs. Mandated Digital Payments in Eswatini," *International Tax and Public Finance*, 2026.

Slemrod, Joel, "Tax Compliance and Enforcement," *Journal of Economic Literature*, 2019, *57* (4), 904–954.

StataCorp, *cmxtmixlogit: Panel-data Mixed Logit Choice Model* Stata Press 2025.

__ , *cmxtmixlogit postestimation: Postestimation tools for cmxtmixlogit* Stata Press n.d. Online manual, accessed 8 November 2026.

van Dijke, Marius and Peter Verboon, "Trust in Authorities as a Boundary Condition to Procedural Fairness Effects on Tax Compliance," *Journal of Economic Psychology*, 2010, *31* (1), 80–91.

Wiswall, Matthew and Basit Zafar, "Preference for the Workplace, Investment in Human Capital, and Gender," Staff Report 767, Federal Reserve Bank of New York 2017. Revised March 2017 version.

__ **and** __ , "Preference for the Workplace, Investment in Human Capital, and Gender," *The Quarterly Journal of Economics*, 2018, *133* (1), 457–507.

World Bank, "Property Tax Diagnostic Manual," Technical Report, World Bank, Washington, DC 2020. Case study on Punjab Province, Pakistan.

A Conceptual comparisons and restrictions

Let H_0 denote an inspector without an informal payment, H_1 an inspector offering a concession for an informal payment, and S satellite assessment. In monetary units, the systematic valuation is

$$V_{ij} = -P_{ij} + h_i(e_{ij}) + \beta_i(e_{ij})\mathbf{1}\{j = S\} - m_i(e_{ij})\mathbf{1}\{j = H_1\},$$

where e records whether the initial bill is correct, excessive or understated, $h_i(0) = 0$, $\beta_i(e)$ values satellite relative to an inspector without a concession, and $m_i(e)$ is the noncash cost of accepting the concession. Negative $m_i(e)$ denotes a positive concession valuation.

With the same initial bill, an inspector concession is preferred to an inspector without a concession when the expenditure saving is at least $m_i(e)$, and to satellite when it is at least $\beta_i(e) + m_i(e)$. If the initial bills differ, their $h_i(e)$ terms also enter. Comparing two inspector concessions at equal expenditure gives

$$V_{H_1}(P, e) - V_{H_1}(P, 0) = h_i(e) - [m_i(e) - m_i(0)].$$

The fixed accuracy comparisons therefore combine initial accuracy and the concession valuation associated with the initial bill. Separating these terms uses the full receipt variation and the maintained analysis model.

Under equation (10), $\beta_i(0) = -1000\alpha_i/b_p$ and $-m_i(0) = -1000\kappa_i/b_p$. The conceptual checks are: C1, savings alone, requires zero mean concession valuation for correct, excessive and understated initial bills ($\bar{\kappa} = \lambda_O = \lambda_U = 0$); C2, concession valuation unchanged across these initial bills, requires $\lambda_O = \lambda_U = 0$; C3 predicts a lower required saving after overassessment ($\lambda_O > 0$); and C4 predicts greater relative inspector valuation after overassessment ($\delta_O < 0$). C3 and C4 are directional hypotheses. Savings alone for every taxpayer would additionally require no characteristic-related or residual concession heterogeneity. Table E.3 tests the mean restrictions and compares λ_O directly with λ_U.

For Table 5, write $s_i = fQ_i$ and set initial-error interactions to zero at correct bills. The satellite-minus-inspector indices are

$$d_i^A(e, f) = \alpha_i - \kappa_i + \delta_e - \lambda_e + b_p f Q_i/1000 \pm \pi,$$
$$d_i^B(e, f) = \alpha_i - \kappa_i - \gamma_e - \lambda_e + b_p f Q_i/1000 \pm \pi.$$

Panel A's common initial-error penalty cancels. Panel B's satellite bill is correct, so its initial-error interaction is zero and the inspector's initial-error term remains. Each probability integrates the logistic function of this index over persistent preferences and balances the two receipt orders. The crossing of the resulting population-average probability includes choice shocks; the individual deterministic saving thresholds above concern systematic valuations.

B Survey, Sampling and Institutional Context

B.1 Sampling frame and recruitment

The survey file contains 6,088 approached properties, of which 2,100 have a completed interview.

Using this sampling frame, for the taxpayer survey, interview targets were specified by tax circle, locality and administrative land-area category. Eligible properties were grouped into four land-area bands: 1–5 marla, above 5 through 10 marla, above 10 through 20 marla, and above 20 through 40 marla. The overall target of 2,100 interviews was allocated across localities in proportion to their aggregate recorded covered area. Within each locality, targets were apportioned across land-area-by-storey cells in proportion to their eligible property counts. Properties were randomly ordered within each cell to construct primary and reserve lists. Out of the total sampling frame of 10,564 properties, fieldwork yielded 2,100 completed interviews from 6,088 approached properties. Table 2 summarizes the survey sample.

The recorded receipts provide reference liability Q, initial bill I, final official payment T, informal payment B and expenditure P. Reconstruction uses displayed amounts, including the amount replacing a struck-through figure, and verifies $P = T + B$, a common positive Q within each pair and zero satellite informal payment. The study design comprises 21,000 paired tasks from 2,100 respondents, with two alternatives per task. Each retained pair has one selected alternative; there are no duplicate respondent–task records.

All 2,100 respondents complete ten paired tasks: four fixed tasks followed by six adaptive tasks. Four inspector receipts show a small official reduction without an informal payment; these remain coded without a concession. Initial error is the sign of $I - Q$.

The four descriptive comparisons retain 2,100, 2,033, 2,060 and 1,624 respondents, respectively. Eligibility requires the displayed attributes to match the stated comparison: equal correct bills and expenditure for the assessor comparison; a strict expenditure saving for the two-inspector concession comparison; almost identical payment for the third comparison; and equal expenditure and informal payment for the fourth. These restrictions apply to the descriptive shares only. Every completed respondent remains in full-sequence estimation. Satellite is shown first in the first comparison, the concession second in the second, and the correct bill first in the last two.

For 36 respondents flagged as exempt, displayed Q is hypothetical: Rs 5,000 for 35 respondents and Rs 9,000 for one. The full sample's displayed reference liability ranges from Rs 1,000 to Rs 276,000; it is held constant within respondent. This benchmark is kept distinct from actual liability.

The choice outcome is coded one for the selected receipt and zero for the competing receipt. Satellite assessment is coded $S = 1$ (inspector $S = 0$); an informal-payment concession is coded $D = 1$ only when $B > 0$. The initial bill is too high when $I > Q$, too low when $I < Q$, and correct when $I = Q$; the two error indicators are mutually exclusive. Expenditure enters as $(T + B)/1000$. The receipt-position indicator is one for the second displayed offer and zero for the first. Assessor-by-error and concession-by-error terms are products of these indicators.

B.2 Survey wording and response codes

Table B.1 consolidates the questionnaire wording, substantive response scales and reported comparison endpoints. Panels A–B cover all ten substantive characteristics in the common model; Panel C gives its three social-desirability items. Panel D documents additional survey measures. English wording follows the fielded instrument, with typographic standardisation.

Code 97 denotes "Don't know" and code 98 "Refused to answer"; both are treated as missing, together with blanks and values outside the stated substantive range. Code 77 in S9.2 records a

substantive "Other" response. Ordinal variables in the common model enter as their numerical scores, so the stated contrasts use the coded distances between response categories.

Table B.1: Survey wording, response codes and analysis comparisons

Measure and item	English questionnaire wording	Responses and analysis coding
A. Institutional expectations and beliefs		
Officer trust (S3.9)	In general, how much do you trust property-tax officers to value properties fairly?	1 = Not at all; 2 = A little; 3 = Mostly; 4 = Completely. Reported comparison: 2 → 3.
Expected correction (S5.5)	If a computer assessment of your property were wrong, how easy would it be to get it corrected?	1 = Very hard; 2 = Hard; 3 = Easy; 4 = Very easy. Reported comparison: 1 → 3.
Satellite fairness (S5.4)	I would trust a satellite-based property valuation system to treat my property the same as any other, without favouritism.	1 = Strongly disagree; 2 = Disagree; 3 = Agree; 4 = Strongly agree. Reported comparison: 2 → 3.
Informal-payment prevalence (S9.9)	In general, how common is it for property tax officers to ask for or accept informal payments during property assessments in this area?	1 = Never; 2 = Rarely; 3 = Sometimes; 4 = Often; 5 = Very often. Reported comparison: 2 → 4.
Expected satellite accuracy (S5.2)	Imagine 10 houses in your neighbourhood similar to yours. A computer measures each one's covered area from satellite pictures, without anyone visiting. Out of these 10, for how many would the computer get the size roughly right – within about one marla?	Integer from 0 to 10. Reported comparison: 4 → 6 houses.
B. Taxpayer circumstances		
Online-form comfort (S1.6)	How comfortable are you with filling a form online?	1 = Not at all comfortable; 2 = A little comfortable; 3 = Comfortable; 4 = Very comfortable. Reported comparison: 2 → 3.
Administrative contact (S2.1)	In the last year, have you or anyone in your household had any contact with the Excise & Taxation Department regarding your property?	1 = Yes; 0 = No. Reported comparison: 0 → 1. The question specifies the last year.

Continued on next page

Table B.1: Survey wording, response codes and analysis comparisons (continued)

Measure and item	English questionnaire wording	Responses and analysis coding
Property title (S3.14)	In official records, how is this property currently held?	1 = Solely in my name; 2 = Jointly owned / shared title; 3 = In another (family) member's name (transfer/intiqal not done); 4 = Ownership is disputed / under litigation; 5 = No formal ownership record exists. Non-sole title equals 1 for codes 2–5 and 0 for code 1. Reported comparison: sole → non-sole.
Document possession (S3.16a–c)	For this property, which of these do you currently have AND could produce within about a week? (Read each document) Items: registry/sale deed; architectural drawings recording official land and covered area; latest property-tax receipt/PT-1.	Each item: 1 = Yes; 0 = No. Documentation is the mean of valid item responses, on a 0–1 scale. Reported comparison: $1/3 \rightarrow 1$.
Displayed reference liability	Reference liability Q_i displayed on the choice receipts.	Measured in rupees; the model uses $\log Q_i$. Reported comparison: Rs 5,000 → Rs 25,000. For exempt respondents, this is the displayed hypothetical benchmark.
C. Social-desirability items included in the common model		
Courtesy (S8.1)	I am always courteous, even to disagreeable people.	1 = True; 0 = False. Score contribution: the recorded binary response.
Admitting mistakes (S8.4)	I am always willing to admit when I make a mistake.	1 = True; 0 = False. Score contribution: the recorded binary response.
Resentment (S8.5)	I sometimes feel resentful when I don't get my way.	1 = True; 0 = False. Score contribution: one minus the recorded binary response.
D. Additional survey measures		
Document access (S3.17)	When you need them for official work, how easy or difficult is it to find these documents for this property?	1 = Very difficult; 2 = Difficult; 3 = Somewhat difficult; 4 = Easy; 5 = Very easy. A separate question from document possession.
Reassessment communication (S3.6)	In the last six months, have you received any communication from E&T about a change in assessment – informally or formally through a letter?	1 = Yes; 0 = No.

Continued on next page

Table B.1: Survey wording, response codes and analysis comparisons (continued)

Measure and item	English questionnaire wording	Responses and analysis coding
Notice accuracy (S3.7)	If yes, is the new assessed value accurate?	1 = Much too low; 2 = Somewhat too low; 3 = About right; 4 = Somewhat too high; 5 = Much too high. Asked if S3.6 = 1.
Correction intention (S9.1)	In the next 3 months, how likely are you to apply to check or correct this house's tax record – out of 10?	Integer from 0 to 10.
Reason for not contesting (S9.2)	If you do NOT plan to contest, what is the main reason?	1 = Assessment seems fair; 2 = Too costly to appeal; 3 = Don't know how to appeal; 4 = Afraid of consequences; 5 = Too much hassle; 77 = Other (please specify). Asked when S9.1 is 0–4; code 77 is a substantive other category.
Property-information purchase (S6.1)	You can receive the computer's estimated details for your own house (covered area, plot outline). The price would be taken from your Rs 2,000 survey payment. Would you like to buy this for Rs ⟨offered price⟩?	1 = Yes; 0 = No. The offer follows the choice tasks; recorded prices are Rs 0, Rs 100 and Rs 200. Acceptance is coded 1 for Yes and 0 for No.

B.3 Constructed measures and survey timing

Documentation and title. The documentation measure averages the valid binary responses to S3.16a–c. A respondent with partial item responses remains eligible for this measure; all three missing responses give a missing score. This construction uses possession of the documents and ability to produce them within about a week. S3.17 separately asks how easy they are to find. Non-sole title groups jointly held property, title in another family member's name, disputed ownership and absence of a formal record; sole title is the reference category.

Social desirability. Let c_i, a_i and r_i be the binary answers to courtesy, admitting mistakes and resentment. The included score is $c_i + a_i + (1 - r_i)$, ranges from zero to three, and requires all three items to have valid binary answers. It enters through interactions with the satellite and informal-payment-concession indicators, alongside the ten substantive characteristics. Table 6 reports the substantive comparisons.

Reference liability, centring and missing responses. The reference-liability characteristic is the natural logarithm of each respondent's displayed Q_i. For each included characteristic, missing values receive that characteristic's mean among respondents with valid values; the resulting variable is then centred over all 2,100 respondents. These centred characteristics interact with

the satellite and concession indicators. The model includes neither survey-arm nor form-version controls nor missing-response indicators.

Timing and routing. The respondent characteristics and social-desirability items precede the choice sequence; satellite accuracy, fairness and correction beliefs follow the information script. Administrative contact in S2.1 concerns the last year. S3.6 is a separate reassessment-communication question. The property-information purchase offer follows the choice tasks. Routing conditions for the additional items appear in Table B.1.

B.4 Administrative organisation

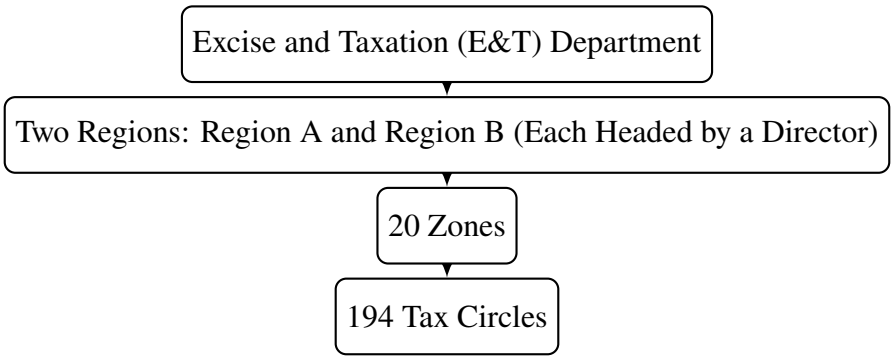


Figure B.1: Flowchart of Property Tax Administration in Lahore

B.5 Spatial coverage for survey sampling frame

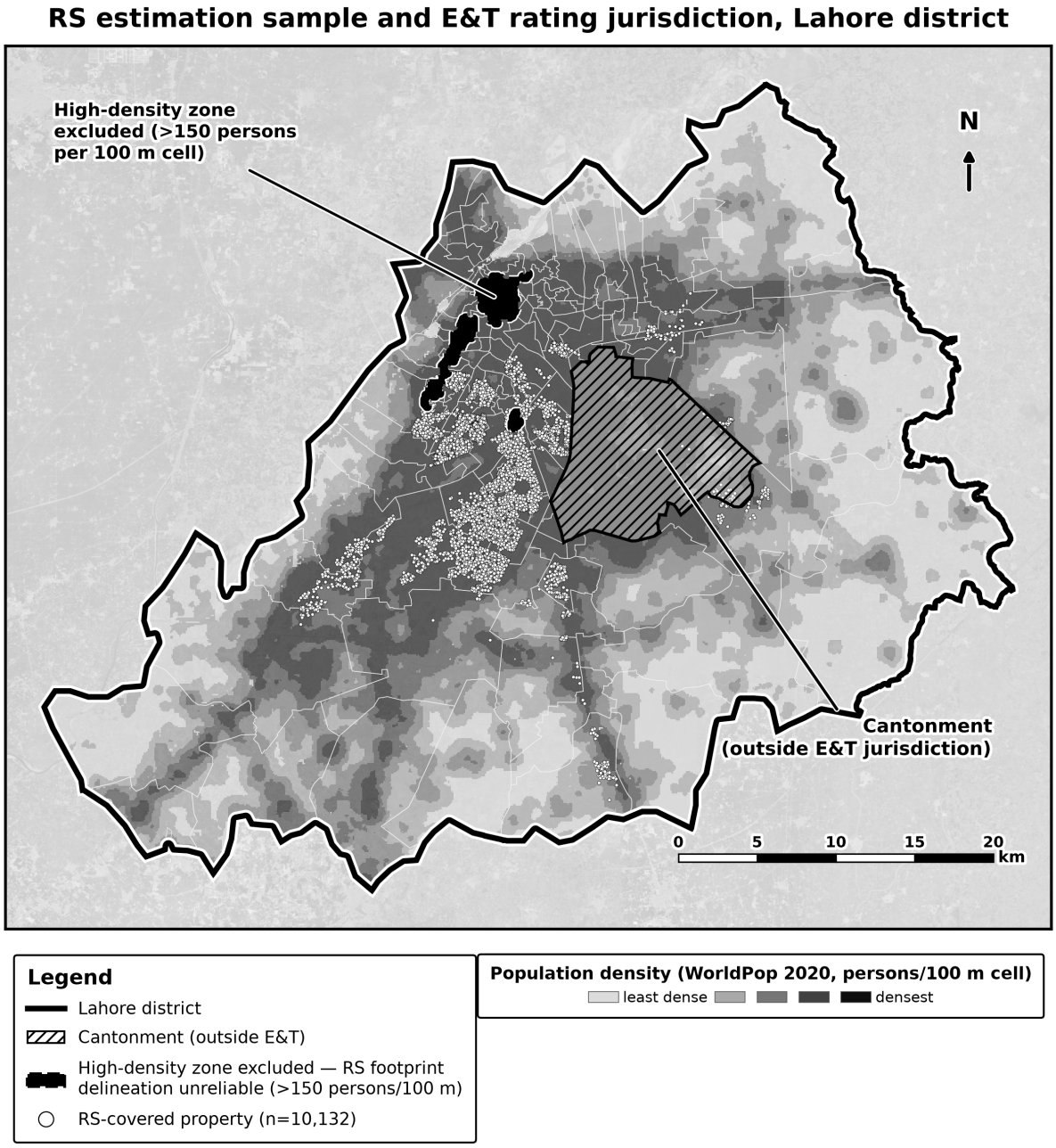


Figure B.2: The remote-sensing estimation sample within Lahore district. Grayscale shading is the WorldPop 2020 population surface (denser cells darker); thin lines are E&T rating jurisdiction. The Lahore Cantonment (hatched) is rated by the Cantonment Board and lies outside E&T jurisdiction; the dashed zone marks the high-density old-city fabric (above 150 persons per 100 m cell) where footprint delineation from imagery is unreliable. Open circles mark RS-covered properties. The sampling frame contains 10,564 properties; the retained map legend reports 10,132. This inherited difference in map and frame coverage remains unreconciled.

C Adaptive choice design

BACE selected informative receipt pairs using earlier choices (Drake et al., 2025b). The field records contain four common opening tasks followed by six adaptive tasks. The study comprises 21,000 paired tasks from 2,100 respondents.

Recovered utility and priors. The available citizen configuration normalises expenditure as $c = 100(T + B)/Q$, signed initial error as $e = 100(I - Q)/Q$, and overassessment as $o = \max(e, 0)/15$. It specifies

$$u = -c + \beta S - m_0 D + m_1 D o, \qquad \Pr(B \text{ chosen}) = \Lambda\{\mu(u_B - u_A)\}.$$

Here β is satellite valuation, m_0 is the concession cost at a correct bill, and m_1 is the reduction in that cost at 15% overassessment. All three are in percentage points of reference liability; μ governs choice consistency. Rupee valuations multiply by $Q/100$; the common scale μ cancels in the monetary ratio.

Table C.1: Prior settings in the recovered citizen configuration

Parameter	Distribution	Support
Satellite valuation β	$N(-0.2, 2^2)$	$\mathbb{R}$
Correct-bill concession cost m_0	$N(5, 5^2)$	$\mathbb{R}$
Cost reduction at 15% excessive vs correct bill m_1	$N(-0.2, 5^2)$	$\mathbb{R}$
Choice-consistency scale μ	$U[0.65, 1.65]$	$[0.65, 1.65]$

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. Satellite valuation is relative to an inspector without a concession. The first three parameters are in percentage points of reference liability; m_0 is a cost and m_1 reduces it at a 15% excessive bill. The scale μ is unitless. Normal distributions are written as $N(\text{mean}, \text{variance})$. These settings combine the recovered configuration with the available citizen parameter file.

Receipt restrictions and updating. The recovered candidate design uses initial errors from −15% to 15% and informal payments from zero to 30% of reference liability, on five-point grids. Satellite informal payment is zero. For an inspector, official reduction equals KB, with K drawn between 1.2 and 2.5. Display rounding and recorded deviations make realised menus differ from these candidate settings; the saved receipts determine the estimation attributes.

After an answer, Bayesian updating multiplies the previous parameter density by that choice's likelihood and normalises it. An information criterion then compares candidate pairs by their expected reduction in uncertainty about individual preferences (Drake et al., 2025b). The frontend requests the subsequent design using the previous choice.

The elicitation utility normalises expenditure and allows concession valuation to vary only with overassessment. The analysis instead estimates expenditure sensitivity and separates accuracy, assessor identity and concession valuation under both excessive and understated initial bills (equation (10)).

Survey receipt screens. Figure C.1 shows an illustrative receipt pair from the design documentation. Figures C.2–C.5 reproduce the four common opening screens, preserving their wording and receipt order. The Rs 20,000 reference liability shown here is illustrative; the analysis reconstructs each respondent's displayed amounts.



Below are two possible assessment situations for your own house. They differ in who assesses the house, how accurate it is, how much lower than the correct amount the recorded bill is, and any informal payment involved.

Please choose the situation you would prefer for your own house.

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	An inspector visits & measures	A satellite checks automatically
Measuring mistake (this time)	+ Rs 1,000 recorded too HIGH (by mistake)	Correct amount (no mistake)
Bill after the mistake	Rs 21,000	Rs 20,000
The deal (chai-pani)	Pay Rs 5,000 and the inspector records Rs 11,900 instead of Rs 21,000	No deal, no one to bargain with
Informal payment	Rs 5,000	Rs 0
Final recorded bill	Rs 11,900	Rs 20,000
What actually leaves your pocket	Rs 11,900 + Rs 5,000 **Rs 16,900**	**Rs 20,000**

Choose A | Choose B

Figure C.1: Illustrative receipt pair: an inspector concession and satellite assessment

Notes: The receipts distinguish reference liability, initial bill, final official payment, informal payment and total expenditure.

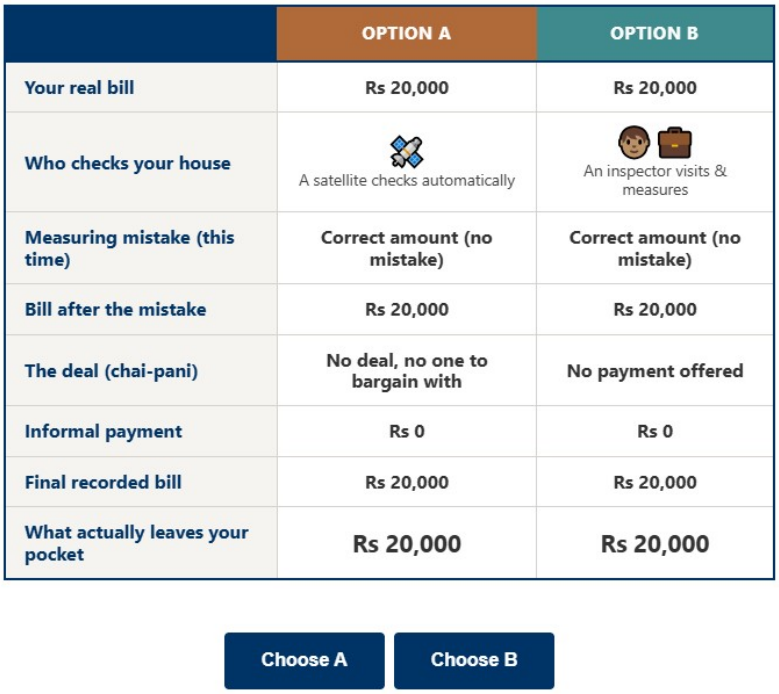


Figure C.2: Scenario 1: satellite and inspector at equal correct bills

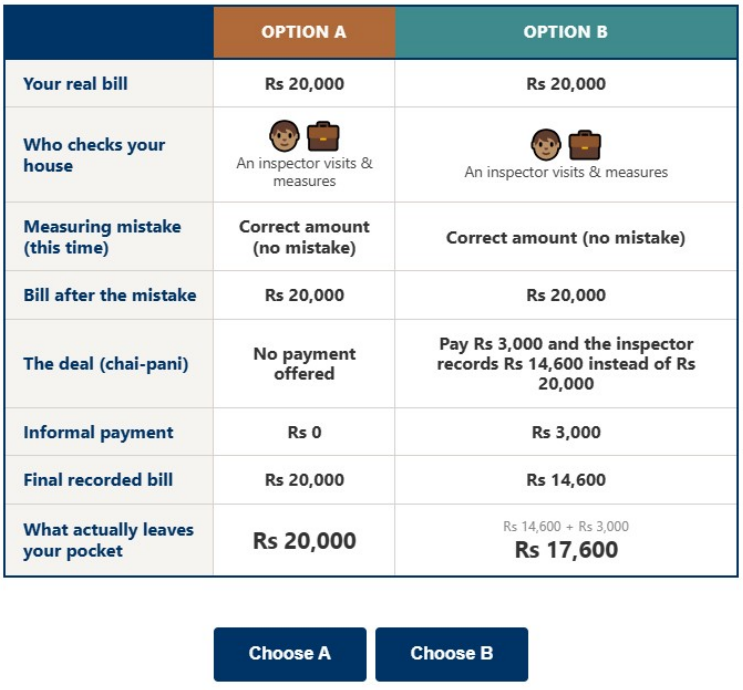


Figure C.3: Scenario 2: inspector offers with and without an informal-payment concession

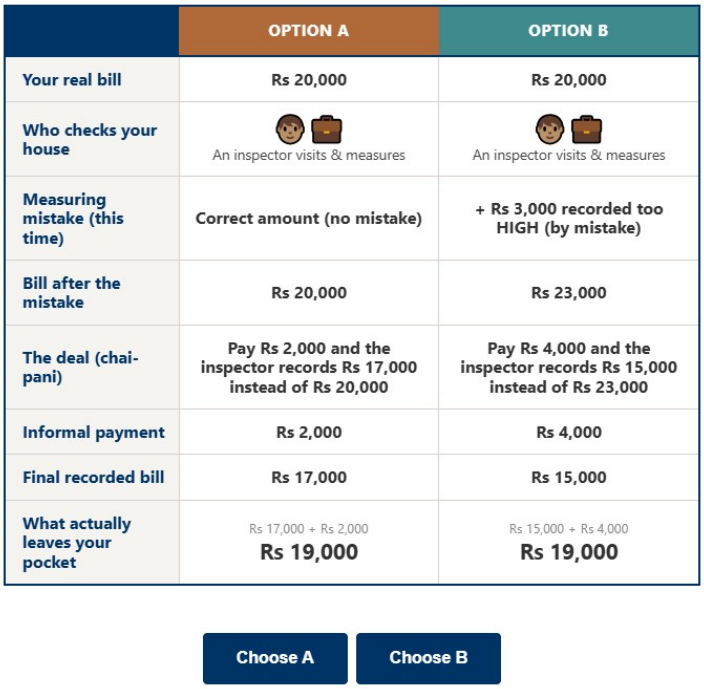


Figure C.4: Scenario 3: correct and excessive initial bills with almost identical payment

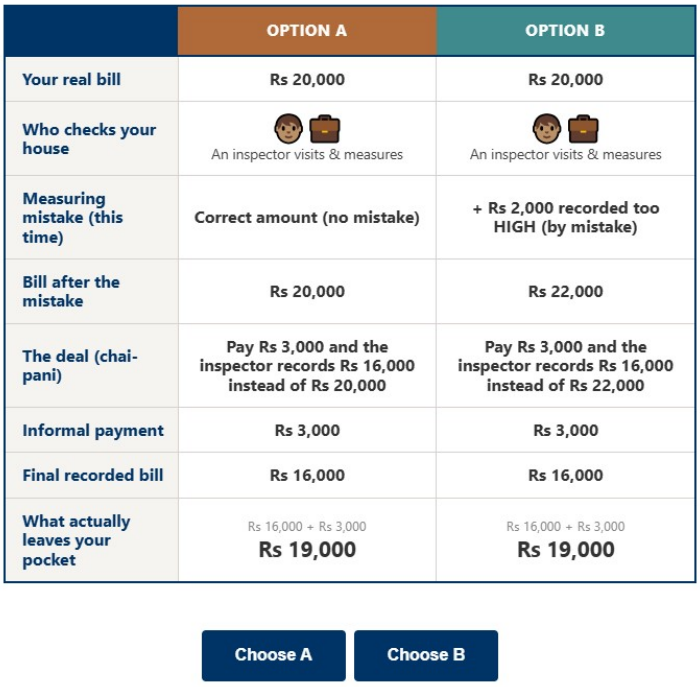


Figure C.5: Scenario 4: correct and excessive initial bills at equal expenditure and informal payment

C.1 Estimation and Inference

Choice probabilities and persistent preferences. For given preference coefficients and type-I extreme-value choice shocks, the probability of choosing offer j is

$$p_{ijt} \equiv \Pr(y_{ijt} = 1 \mid X_{it}, \alpha_i, \kappa_i) = \frac{\exp(v_{ijt})}{\sum_{k \in \mathcal{J}_{it}} \exp(v_{ikt})}, \qquad j \in \mathcal{J}_{it}, \tag{C.1}$$

where X_{it} contains the attributes of both offers. A larger index difference makes an offer more likely to be chosen. The coefficients are in conditional logit units: their signs describe the direction of the conditional valuation, while monetary and probability transformations give the economic magnitudes.

The respondent coefficients persist across tasks:

$$\alpha_i = \bar{\alpha} + x_i'\eta_S + \sigma_\alpha z_{\alpha i}, \qquad \kappa_i = \bar{\kappa} + x_i'\eta_D + \sigma_\kappa z_{\kappa i}.$$

Here x_i contains the respondent-centred characteristics defined in Section 5.3. The two z terms are jointly standard normal with correlation ρ. The expenditure coefficient and initial-error interactions are common. Centring makes $\bar{\alpha}$ and $\bar{\kappa}$ the sample-average conditional means; predictions still average over all respondents and integrate over their residual tastes. The specification allows the association between assessor and concession preferences to be estimated within the common panel model (StataCorp, 2025).

Panel likelihood and adaptive assignment. The panel likelihood first multiplies the conditional probabilities of all choices by a taxpayer and then integrates over that taxpayer's two persistent random coefficients. Numerical quadrature approximates this integral in the panel likelihood. The saved covariance combines the numerical Hessian and respondent likelihood scores in a sandwich estimator. This integration order follows the panel formulation in StataCorp (2025).

Earlier answers influence the choice pairs presented later. Estimation conditions on the realised sequential design: the assignment rule may depend on recorded history but must be independent of the current unobserved choice shock, and its probability must contain no additional unknown preference parameters conditional on that history. Under these maintained assumptions, the assignment factors can be omitted from the likelihood for preference parameters. Preference heterogeneity is persistent and choice shocks are independent across tasks conditional on the coefficients and choice pairs. The estimates describe stated choices under these assumptions (Drake et al., 2025b).

Statistical inference. The observed fixed-task shares use Wilson 95% intervals and task-specific receipt eligibility. Model probability contrasts and expected fiscal allocations use the full covariance matrix and delta-method 95% intervals. Monetary coefficient ratios use Fieller intervals, with delta-method standard errors and two-sided tests also retained. These calculations propagate parameter uncertainty conditional on the reference profiles. Section 5.2 gives the monetary transformations; Section 5.3 defines the multiple-testing families.

Population predictions and reference scenarios. Population probabilities integrate equation (C.1) over the fitted random-coefficient distribution for each respondent, then average respondents equally. Each receipt order receives weight one-half. Categorical contrasts change the specified receipt attributes and recompute all interactions. The probability calculations follow these alternative-specific prediction definitions (StataCorp, n.d.).

The central comparison defines net inspector saving as $s_i = P_{iS} - P_{iH}$, including informal payment, and writes $f = s_i/Q_i$. Both scenario panels in Table 5 use each respondent's displayed Q_i. Informal payment is fixed at $0.2Q_i$; changing the saving changes only the inspector's official payment. Panel A gives both offers the same initial bill, while Panel B keeps satellite correct and varies the inspector's initial bill. Contrasts compare the integrated probabilities directly, using their full parameter covariance. The reported 50% crossing solves for the saving at which the population-average probability equals one-half; its confidence set inverts the pointwise probability band within the supported range. Reference receipts are screened against joint paired-receipt support; sparse comparisons are omitted from the main table.

Separating accuracy and concession valuations. The monetary coefficient supplies a common linear benchmark for separating expenditure from the other attributes. The equal-expenditure comparisons between inspector profiles offering concessions for informal payments identify $h(e) - [m(e) - m(0)]$, where $h(e)$ is the monetary value of the initial bill, $h(0) = 0$, and $m(e)$ is the noncash cost of the concession. Separating its components uses the wider receipt variation and the expenditure specification; reductions without an informal payment are rare. Appendix B.2 describes receipt reconstruction.

D Survey-recorded posterior summaries

The survey saved individual posterior summaries following sequential choices. Under the recovered utility in Appendix C, each answer contributes a logistic likelihood; updating combines it with the preceding parameter distribution. Usable respondent-linked means, medians and standard deviations are saved for 2,040 respondents; full draws and updating histories are unavailable.

Table D.1: Survey-recorded individual posterior summaries

Parameter	Mean of posterior means	SD of posterior means	Mean posterior SD
Satellite vs inspector without concession: valuation β	-0.145	2.150	1.446
Inspector concession vs no concession: correct-bill cost m_0	5.842	6.934	2.008
Concession cost: reduction at a 15% excessive vs correct bill m_1	-1.812	3.360	2.276
Choice-consistency scale μ	1.056	0.144	0.265

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. Positive m_0 is a concession cost; positive m_1 reduces that cost at an excessive bill. Each row covers 2,040 respondents. The first numeric column averages individual posterior means; the second is their cross-person standard deviation; the third averages individual posterior standard deviations. The first three parameters are in percentage points of reference liability under the recovered normalisation; μ is the utility scale. These dispersion measures describe different objects from uncertainty about a population mean.

The saved summaries and the primary model use different utility specifications, expenditure normalisations and treatments of individual information. The live estimates combine each respondent's answers with design priors. The primary model estimates a population preference distribution by pooling the full choice sequence, with characteristic-related means and residual heterogeneity. A distribution of individual posterior means also reflects shrinkage and differs from a population distribution of latent preferences (Drake et al., 2025c).

Under the recovered monetary normalisation, multiplying each respondent's saved mean by their own $Q_i/100$ gives an average satellite valuation of Rs -95 and correct-bill concession valuation of Rs -961; the concession conversion uses $-m_{0i}$. Conditioning the common model on the same 2,040 respondents' full recorded histories gives Rs -235 and Rs 32, respectively. The concession valuations differ substantially. The primary analysis estimates accuracy and both concession-state effects separately, alongside expenditure and respondent characteristics; its full-sample population means answer a different question from averages of these saved individual summaries. The main model is used to keep those economic comparisons on one common specification and sample.

E Supplementary Evidence

E.1 Demand for independent property information

Independent property information can help an owner understand a record or question a bill under either assessor. The information offer therefore measures a different demand from the choice of who assesses the property. S6.1 offers computer-estimated property information for Rs 0, Rs 100 or Rs 200, with any price deducted from participation credit. Let Buy_i equal one for acceptance and zero for rejection, and let q_i be the recorded offer price in rupees. I use the 2,099 substantive responses.

I compare acceptance at each positive price with acceptance under free provision using

$$100\text{Buy}_i = \alpha_B + \tau_{100}\mathbf{1}\{q_i = 100\} + \tau_{200}\mathbf{1}\{q_i = 200\} + \varepsilon_i. \tag{E.1}$$

The intercept α_B is acceptance at the free offer, in percentage points. Each τ is the difference at the stated price relative to free provision. A negative coefficient means lower acceptance at that price. This saturated specification estimates the two differences separately rather than imposing a linear response to price. HC3 standard errors allow different variances across groups. The estimates describe differences across the recorded price groups.

Figure E.1 reports acceptance of 94.8% when free, 39.0% at Rs 100 and 39.4% at Rs 200. The largest difference is between free provision and either positive price. The fitted differences are −55.7 and −55.4 percentage points, each with an HC3 standard error of approximately two points (Table E.1). Acceptance is similar at the two positive prices. Independent information attracts broad interest when free, while a substantial share of owners also accept it when payment is required.

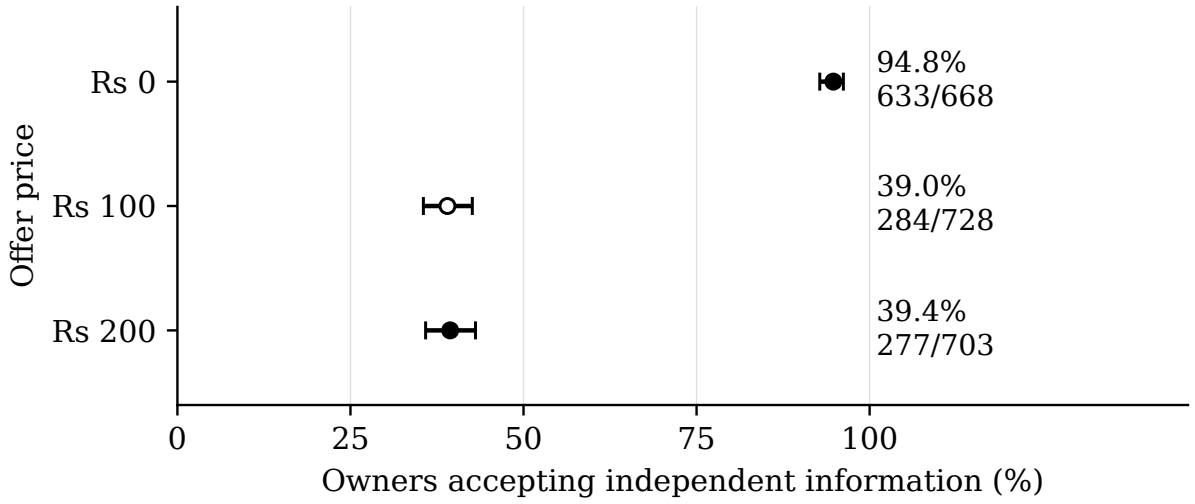


Figure E.1: Acceptance of independent property information by offer price

Notes: S6.1 offers computer-estimated property information, with price deducted from participation credit. There are 2,099 substantive responses; one additional owner at Rs 200 provides no substantive answer. Unweighted shares and Wilson 95% intervals; counts and denominators appear beside the points. The outcome records offer acceptance.

E.1.1 Price-group estimates

Table E.1: Acceptance of property information: paid vs free offers

Offer-price comparison	Estimate (SE)	95% interval
Acceptance when free (%)	94.76*** (0.86)	[93.07, 96.45]
Rs 100 vs free (percentage points)	-55.75*** (2.01)	[-59.68, -51.82]
Rs 200 vs free (percentage points)	-55.36*** (2.04)	[-59.35, -51.36]

Notes: Saturated linear probability model, 2,099 observed responses. HC3 standard errors and two-sided t-reference inference. * p<0.10, ** p<0.05, *** p<0.01. Any positive price is deducted from the survey participation payment. The free-offer constant is the acceptance share; other coefficients compare acceptance at each positive price with acceptance under free provision.

The posted-price comparison retains 668 valid responses at zero, 728 at Rs 100 and 703 at Rs 200. One additional owner in the Rs 200 group supplies no substantive purchase answer. The saturated model reports each paid-price contrast separately. The reported shares show similar acceptance at the two positive prices.

E.2 Choice probabilities and tests of concession valuation

The following tables report probability comparisons and coefficient tests from the same primary model as the monetary valuations. Table E.2 reports changes in selection probability, in percentage points. Positive entries raise selection of the named offer; negative entries lower it.

Table E.2: Changes in selection probability under specified receipts

Comparison	Change (pp)	95% interval	p
Satellite vs an identical inspector offer without concession	0.4	[-1.2, 2.0]	0.641
Inspector concession: Rs 19,000 vs Rs 18,000 total payment	-6.4***	[-7.8, -5.1]	< 0.001

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. The first row reports the satellite choice probability minus the 50% probability under two identical inspector offers. The second reports the concession choice probability at Rs 19,000 minus that at Rs 18,000. Common full-sequence panel mixed logit; 2,100 respondents. The study design comprises 21,000 paired tasks: four fixed and six adaptive tasks per respondent. Inference uses the full respondent-score covariance. Entries are percentage-point changes with pointwise 95% delta-method intervals. The satellite comparison replaces one of two identical inspector offers with satellite; both initial bills are correct, payments are equal and no informal payment is made. The price comparison raises the concession's official payment from Rs 15,000 to Rs 16,000, holding its informal payment at Rs 3,000, against an inspector charging Rs 20,000 without a concession. Both initial bills and reference liabilities are Rs 20,000. Predictions balance receipt order. Tests use the full covariance, are two-sided and unadjusted. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Interpreting the probability comparisons. The first row starts with two identical inspector offers: correct bills, equal expenditure and no informal payment. Each has a 50% choice probability. Replacing one with satellite raises its predicted share to 50.4%, a change of 0.4 percentage points (95% interval: -1.2 to 2.0 points). The model averages individual choice probabilities, so this near-even split can coexist with the slightly negative mean rupee valuation in the main valuation table.

The second row changes money while keeping both assessors as inspectors. One offers an official payment of Rs 15,000 plus Rs 3,000 informally; the other charges Rs 20,000 without an informal payment. Raising the first offer's official payment to Rs 16,000 cuts its net saving from Rs 2,000 to Rs 1,000. Its predicted choice share falls by 6.4 percentage points (95% interval: -7.8 to -5.1 points). Both initial bills remain correct at Rs 20,000.

Table E.3 reports joint Wald statistics for C1–C2 and utility coefficients for the remaining comparisons. A positive C3 coefficient means that overassessment raises the concession premium. A negative C4 coefficient means that overassessment lowers the satellite premium. The final row directly compares the concession shifts under excessive and too-low initial bills.

Table E.3: Tests of concession valuation and initial-bill differences

Test	Wald statistic / coefficient (SE)	Null	p
C1: Mean concession premium is zero at each initial bill	$\chi^2_3 = 17.67^{***}$	$\bar{\kappa} = \lambda_O = \lambda_U = 0$	< 0.001
C2: Same concession premium at correct, excessive and too-low bills	$\chi^2_2 = 13.13^{***}$	$\lambda_O = \lambda_U = 0$	0.001
C3: Concession premium: excessive vs correct initial bill	0.313^{***} (0.110)	$\lambda_O = 0$	0.004
C4: Satellite premium: excessive vs correct initial bill	-0.068 (0.116)	$\delta_O = 0$	0.556
Concession premium: excessive vs too-low initial bill	0.021 (0.138)	$\lambda_O = \lambda_U$	0.877

Notes: A concession is an official-bill reduction in exchange for an informal payment to the inspector. A premium is the utility advantage over an inspector without a concession at the same initial bill and total expenditure. Positive coefficients indicate a larger premium at the first named initial bill than at the second. Common full-sequence panel mixed logit; 2,100 respondents. The study design comprises 21,000 paired tasks: four fixed and six adaptive tasks per respondent. Inference uses the full respondent-score covariance. C1 and C2 are joint Wald tests of the displayed restrictions. Other rows report utility coefficients with standard errors in parentheses. All tests use the full respondent-score covariance and are two-sided; p-values are unadjusted. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What changes with the initial bill? Table E.3 rejects the restrictions that concessions matter only through money saved (C1), and that their noncash valuation is the same after correct, excessive and understated initial bills (C2). Relative to a correct bill, an excessive bill raises the conditional valuation of an inspector concession, so less saving is needed to make that arrangement attractive (C3). The estimated increase in the inspector's relative valuation over satellite following overassessment is statistically insignificant (C4). The direct test of the concession shift under excessive versus too-low bills is also statistically insignificant ($p = 0.877$). Thus an excessive bill raises the concession premium relative to an inspector without one, even though a correct bill is preferred when both offers include concessions.